The University of Hong Kong

FITE4801 Final Year Project – Interim Report

A Decentralized Data Wallet Unlocking Individual Data into the Next Digital Asset

Name: Luk Wing Kei (3033104368)

Wong Tsz Tung (3036065408)

Supervisor: Prof. K.H. Chow

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Abstract

Personal data has emerged as a critical digital asset, yet centralized platforms currently monopolize its control and profits, leaving individuals without oversight or compensation. This interim report presents the development of the Decentralized Data Wallet (DDW), a blockchain-based platform designed to restore personal data sovereignty by enabling the privacy-preserving sharing of contributor-verified insights. Following a Multivocal Literature Review (MLR), the DDW was engineered using a four-layer hierarchical architecture. The project has reached a significant milestone with the delivery of a functional web-based proof-of-concept (PoC). Key features currently operational in the contributor interface include the Data Upload Module, a Token Rewards Earnings Dashboard, secure Wallet Integration, and a Token Redemption module for on-chain value exchange. On the backend, a predictive engine utilizing a Random Forest model has been implemented to synthesize behavioral metadata into actionable Alternative Credit Scores, providing a standardized metric for enterprise data consumers. The DDW offers a practical solution to the privacy-monetization paradox, allowing individuals to profit from high-value, anonymized insights without exposing raw personal data. While current results align with the project schedule, the next phase will focus on refining the insight generation engine and conducting Alpha and Beta testing to evaluate operational stability and usability. Furthermore, interviews with industry practitioners are planned to assess the platform's real-world scalability. Ultimately, this prototype establishes a credible pathway toward an equitable data ecosystem where sovereignty, privacy, and financial reward coexist.

Acknowledgement

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Abbreviations

API	Application Programming Interface
CIDs	Content Identifiers
DAC	Decentralized Access Control
DDW	Decentralized Data Wallet
DID	Decentralized Identifier
DPIs	Decentralized Personal Insights
dApp	Decentralized Application
ETH	Ethereum
EVM	Ethereum Virtual Machine
IPFS	InterPlanetary File System
KNN	K-Nearest Neighbors
MLR	Multivocal Literature Review
NFT	Non-fungible token
OCR	Optical Character Recognition
PoC	Proof-of-Concept
PoS	Proof-of-Stake
P2P	Peer-to-Peer
SLR	Systematic Literature Review
SBTs	Soulbound Tokens
UI	User Interface
UCD	User-centered design
ZKPs	Zero-knowledge Proofs

1. Introduction

1.1 Project Background

Personal data has become one of the most valuable digital assets in today's digital economy [1]. The World Economic Forum [2] claimed that personal data has emerged as a new asset class and individuals should have the right to capitalize their personal data. In the traditional centralized data ecosystem, tech giants and platforms aggregate individuals' data through proprietary infrastructures, user accounts, and behavioral tracking systems. This data, stored on centralized servers, is analyzed for patterns, and monetized through product optimization and third-party data sales, often without users' informed consent [3],[4]. This centralized approach results in two main negative outcomes: first, severe information asymmetries between large corporations and individuals. Corporations know almost everything about the user, meanwhile, individuals have very limited visibility or control over what data is being collected, how it is analysed or shared, and for what purposes it is used [4]; second, it perpetuates economic inequity, where corporations become richer by exploiting user data to generate trillions in annual revenues, while users remains financially uncompensated from contributing their personal data for enterprises' revenue growth [1]. Data silos worsen this inequity, as centralized institutions hoard data within closed ecosystems to prevent data from being shared or used in ways that might benefit individuals or society [5].

In recent years, blockchain-powered decentralized solutions have emerged as ideal approaches to the two aforementioned challenges. They enable grant self-sovereign control to data contributors to determine how to use and handle their data and monetization opportunities of personal data as assets. DataMesh+, a blockchain-powered self-sovereign data marketplace proposed by Merlec and In [6], empowers peer-to-peer (P2P) data exchange between data buyers and sellers. Sellers can choose whether to list the encrypted datasets they have on the market, set their own prices for sale, and sell their data directly to data buyers without relying on intermediaries that typically charge high commissions when executing data transactions. Similarly, Ocean Protocol, an Ethereum-based decentralized data marketplace, enables data sellers to monetize off-chain datasets by creating datatokens and selling them to data consumers to give them the right to access off-chain datasets without transferring data ownership. This turns data into revenue streams without ownership transfer and brings the possibility for everyone to monetize their data on the decentralized platform [7]. Thus, the concept of decentralized data ownership and monetization is not just theoretical, as proposed in [6], but is already being implemented in real-life contexts by major companies.

1.2 Project Motivation

However, decentralized data monetization solutions exhibit significant technical limitations. First, data fragmentation across platforms scatters users' personal information, making it difficult for individuals to maintain a unified view and effective control of their data [8]. Second, weak data provenance, which refers to the limited ability to track the origin and history of data, reduces the authenticity and traceability of shared information, leading to inaccurate or unreliable insights generated from low-quality, polluted aggregated datasets [8], [9]. Third, inefficient data interoperability, as most current systems focus on monetizing datasets in .csv formats without supporting diverse data modalities, restricting the range of monetizable data assets, and limiting the generation of multi-dimensional insights [10].

To address these challenges, a Decentralized Data Wallet (DDW) is proposed in this project as a novel system infrastructure to enable individuals to transform their personal data into digital assets in the form of Decentralized Personal Insights (DPIs), represented as tokens. Individuals receive tokens as rewards for uploading useful personal data, which are then used to generate valuable DPIs accessed by enterprises through a peer-to-peer (P2P) network. As data contributors, individuals can upload their personal data, control which enterprises may access their profiles, selectively choose which DPIs to share, and define the time and duration of DPIs access by data consumers. Enterprises can request access to the DPIs in individual data profiles. Upon approval and successful sharing of insights with data consumers, individuals receive tokens as rewards. The reward amount is determined by the type, relevance, and utility of the shared insights. This token-based compensation system ensures that individuals are economically incentivized for their data contributions, while enterprises are restricted only to access the anonymized DPIs, rather than raw data, to preserve individual data privacy.

The central motivation of this project arises from the need to generate accurate, high-quality insights directly from individual data, rather than relying on aggregated or intermediated datasets. To achieve this, a token-based incentive mechanism is introduced to reward participants based on the value generated from their personal data, encouraging individuals to continuously enrich and maintain robust and reliable data profiles. Simultaneously, enterprises, in return, gain access to ethically sourced and verifiable insights derived directly from authenticated users' data profiles, rather than through intermediated data aggregators. This mechanism strengthens analytical reliability across the entire data value chain - from data collection and verification to insight generation and enterprise application - thereby supporting more informed business decision-making and sustainable growth. Thus, dual focus, which empower individuals to contribute high-quality data while providing enterprises with reliable, quality insights, forms the central motivation for the development of the project.

Current decentralized data marketplaces prioritize trading raw datasets over their derivable actionable insights, exposing privacy risks in exchanges. Existing systems lack frameworks for tokenizing these insights as standardized and monetizable digital assets. Thus, the DDW bridges the gap by enabling privacy-preserving insight monetization for individuals and delivering verifiable and high-quality information to enterprises. Future work is hoped to integrate the DDW with existing decentralized infrastructures and regulatory frameworks to enhance its interoperability with the real-word context . Additionally, it aims to establish a scalable reference architecture for decentralized data sharing and monetization, providing a foundation for further research in this emerging domain.

1.3 Project Objective

The primary goal of this project is to conceptualize, design, and implement a Decentralized Data Wallet (DDW) that bridges the gap between secure data ownership and machine learning analytics through a tokenized reward system. The specific objectives are as follows:

- To design a functional system architecture that integrates machine learning with blockchain technology to transform raw personal data into tokenized and monetizable digital assets known DPIs.

- To safeguard data sovereignty and data contributors' privacy by implementing relevant privacy-preserving protocols, ensuring users maintain authority over access to grant or deny access requests to their DPIs from data consumers while providing enterprises with verified, anonymized insights instead of raw data.

- To enhance the accountability of the analytical pipeline by establishing a blockchain-based framework for data provenance and auditability, ensuring that the origin and integrity of data inputs are verifiable for enterprise applications .

- To develop an equitable data monetization model that rewards contributors with tokens based on the value, freshness, and analytical demand of their data, incentivizing the maintenance of comprehensive and holistic data profiles.

Together, these objectives aim to establish a transparent data-to-insight pipeline that unlocks the potential of personal data as a monetizable digital asset. By utilizing blockchain to ensure data supply chain auditability and a token-based framework for fair financial recognition, the project bridges the gap between secure data ownership and the strategic benefits accessible to participants in the digital economy.

1.4 Project Scope

A DDW is engineered as a versatile infrastructure capable of supporting a wide array of personal data types and modalities. To ensure technical feasibility and a focused evaluation during the initial development cycle, a preliminary scope has been established. This scope is intended to be dynamic; it will be adjusted tentatively according to the prototype development schedule, specific requirements from supervisors, and practical limitations encountered during implementation. As the project progresses, the scope may expand or shift to accommodate more robust machine learning (ML) requirements and data availability.

The project scope for the DDW is delineated across three dimensions: targeted data domains (Section 1.4.1), application scenarios and use cases (Section 1.4.2), and technical modalities (Section 1.4.3). This structured framework ensures the system's versatility is validated through high-value use cases while maintaining the iterative flexibility required for software engineering and machine learning optimization.

1.4.1 Data Domains

In the current phase, the project focuses on establishing a functional framework within specific data domains to demonstrate the DDW's utility and applications. While travel-related behavioral data, such as airline choices, accommodation history, and overseas dining records, has been selected as the primary data substrate to validate the DDW application's core functionality due to its high analytical value for financial applications, the system is designed to be domain-agnostic. The scope may be adjusted to accommodate a greater variety of data domains depending on the feature selection process required for training the system's underlying machine learning algorithms. This adaptability ensures that the DDW can ingest and process diverse datasets that offer high relevance to the chosen application scenarios.

1.4.2 Application Scenarios and Use Cases

The primary application scenario for this project is the generation of DPIs for alternative credit scoring. This involves leveraging non-traditional data sources to assess an individual's creditworthiness, thereby promoting more inclusive lending decisions and reducing reliance on centralized financial histories [11]-[14]. While alternative credit scoring remains the primary focus, the architecture is being developed to support additional use cases that may be explored in the subsequent semester. These potential expansions include traditional credit scoring frameworks and health or behavioral wellness applications. The final selection of these applications will be refined in collaboration with project supervisors to ensure the insights generated are both impactful and technically verifiable within a decentralized context, and functional applicability within real-word context.

1.4.3 Supported Data Modalities

To facilitate the generation of high-fidelity, multi-dimensional insights, the DDW integrates four primary data modalities to ensure a comprehensive user profile . First, structured text data is utilized for processing organized datasets like JSON or CSV files. Second, time-series data is incorporated to capture and analyze behavioral trends and patterns over specific durations. Third, image data is supported to allow the system to ingest visual information, which can be processed through optical character recognition (OCR) techniques to extract relevant features for the DPI engine. Finally, PDF documents are included to support the automated extraction of data from official reports and formal documentation. The inclusion and priority of these modalities will be adjusted periodically to align with the evolving training data requirements of the machine learning algorithms.

1.5 Project Deliverables

The project delivers three interdependent components. The first deliverable is a document providing a systematic overview of the DDW's system architecture and protocol design, detailing its end-to-end operational flow. The second deliverable is a fully functional web-based DDW application prototype, accompanied by a proof-of-concept (PoC) validation report. The application includes a front-end user interface for data contributors and consumers and a backend that handles the system's core operations. It serves as a working prototype to demonstrate the functionality of DDW. The PoC validation report documents the technical correctness and operational stability from alpha testing results, while system performance metrics, user feedback, and the iterative refinements made to the DDW's architecture to enhance its system performance and user experience will be documented based on the beta testing results. The third deliverable is an evaluative discussion report analysing the practicality and limitations of the DDW in alternative credit scoring applications. The report presents quantitative metrics to evaluate the DDW's usability and operational feasibility, beta testing participant feedback regarding trust, and user experience with the DDW, and insights from industry practitioners to justify its real-world applicability.

1.6 Report Outline

The remainder of the report is organized as follows: Section 2 outlines the methodology, including multivocal literature review, prototyping and testing. Section 3 presents current progress and limitations. Section 4 concludes the key findings and recommendations.

2. Project Methodology

The project methodology is structured into two subsections. Section 2.1 outlines the development of the technical protocol specification blueprint using a Multivocal Literature Review (MLR). Section 2.2 describes the design of the DDW user interface, the implementation of the backend architecture, and the approach to alpha and beta testing. Section 2.3 concludes the methodology by applying an empirical mixed-methods analysis to assess the DDW's real-world practicality and constraints.

2.1 Multivocal Literature Review

Multivocal Literature Review (MLR), a systematic review approach that integrates both academic research and grey literature, including white papers, technical reports, standards documentation, and practitioner blogs [15], will be adopted to develop the technical protocol blueprint for the DDW. Unlike a traditional Systematic Literature Review (SLR), which relies exclusively on peer-reviewed academic literature, the MLR is particularly valuable in fast-evolving research domains where academic research often lags behind industrial innovation [15].

The core aim of this MLR is to facilitate a deep inquiry into the practical application of blockchain technology across various industries and current industry practices. To direct this scope of inquiry, several thematic research objectives are utilized to strengthen the development of the four core protocol layers and the supporting components of the system architecture. These areas of investigation are intended to be dynamic and without fixed limitation throughout the research process. By synthesizing insights from academic databases and industry repositories, the MLR ensures that system architecture and the protocol design of DDW are both theoretically robust and practically implementable.

Sources will be drawn from academic databases and industry repositories, including insights produced by industrial practitioners, companies' whitepapers, technical reports, and protocols on tokenization projects. Literature published between 2020 and 2025 is prioritized to capture the latest advancements in the decentralized data economy. This comprehensive synthesis ultimately provides a verified foundation for aligning the project's technical specifications with contemporary industry standards and emerging blockchain paradigms.

2.2 Development: Frontend Design, Backend Implementation & Testing

2.2.1 Proof-of-Concept (PoC) Development – DDW Prototype

The development of the DDW prototype follows an evolutionary prototyping methodology, focusing on the functional validation of the system's decentralized core while iteratively refining user interactions. A functional, web-based version of the prototype will be developed to demonstrate end-to-end user journeys for both the data contributors (individuals) and data consumers (enterprises). The User Interface (UI) is meticulously designed to simplify the complexities of blockchain, such as gas management and cryptographic signing, while ensuring complete transparency for all stakeholders.

By focusing on these key interactions (see Figure 1), the PoC serves as a functional testing ground to test the integration of frontend components with the blockchain backend, ensuring the system is both user-centric and technically robust.

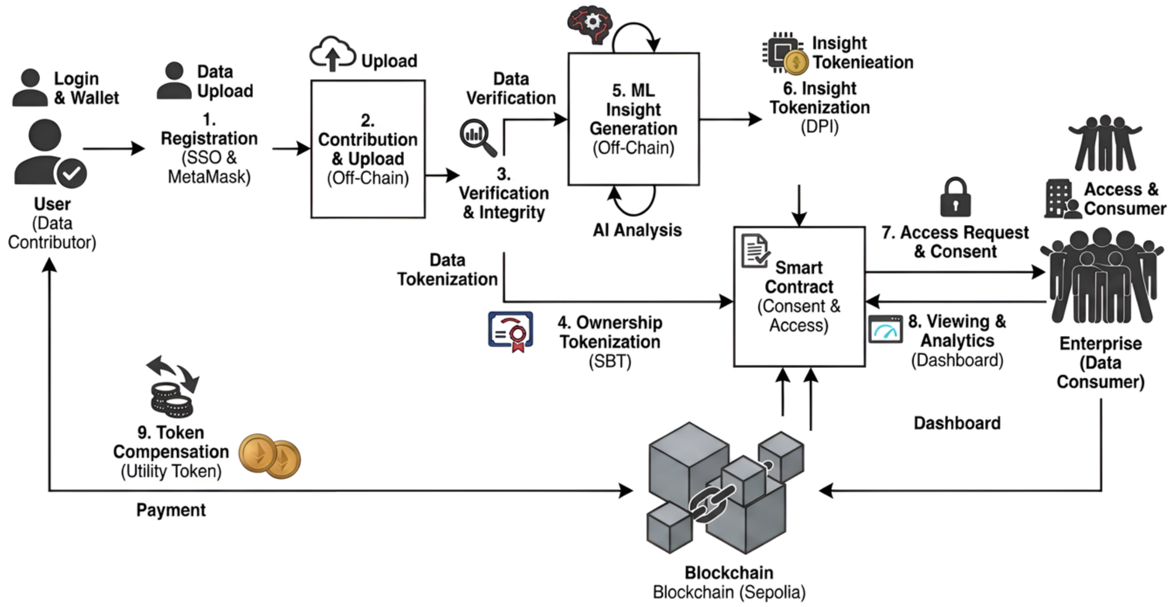

Figure 1 High-Level Workflow of the DDW under a Hybrid Off-Chain ML Model

The Sepolia testnet on Ethereum will be selected as the blockchain platform for building the backend decentralised base layer for DDW [16]. The Ethereum Sepolia testnet is a simulated blockchain environment that mirrors Ethereum's mainnet functionality, using valueless test Ether for free development and testing of smart contracts and tokenization [16]. It is chosen because it provides a cost-effective and robust environment for developing and testing smart contracts and tokenization for our architecture at an initial stage.

Following the prototype, the DDW prototype undergoes a two-phased validation process to ensure technical stability and user acceptance.

2.2.2 Phase I - Internal Technical and Operational Evaluation (Alpha Testing)

The PoC undergoes Alpha Testing to confirm functional integrity prior to public-facing trials. Utilizing a controlled environment with two data contributors and one data consumer, this phase focuses on verifying the system's core decentralized logic. Iterative refinements are implemented based on internal performance data to preempt functional errors during subsequent user testing.

- **Technical Verification**: Assessing the accuracy of the Data-to-Asset Transformation (raw data to DPI tokens), the logic of Smart Contract Execution (consent and compensation), and the efficacy of Privacy-Preserving Mechanisms.
- **Operational Assessment**: Evaluating Layered Architecture Integrity to ensure cohesive interaction between the frontend, blockchain, and off-chain database, while monitoring system reliability under simulated real-time conditions.

Remarks: The internal evaluation criteria outlined above are specifically designed for the Hybrid (Off-Chain ML) model. These criteria are subject to amendment in subsequent project stages as the architecture transitions to a Pure On-Chain Model, where the entire ML engine will be executed within the smart contract environment.

2.2.3 Phase II - User-Centered Design (UCD) Evaluation (Beta Testing)

Following the refinements made during the Alpha testing phase, Beta Testing will be conducted to assess the prototype's utility in real-world scenarios. In this phase, the application will be tested by 20 participants, who will act as both data contributors and a select group of data consumers. Participants will complete a series of representative tasks to evaluate the system's functional performance and overall user experience.

Feedback from the participants will be gathered and analyzed through both quantitative and qualitative metrics (see **Table 1)**. A UCD approach ensures that the system is designed in alignment with user needs, expectations, and practical use cases, enhancing its relevance and usability in real-world conditions.

Feedbacks	Measurement & Method
Quantitative measures	• Task completion rate (how many finished tasks successfully) • Error rate (how many mistakes occurred) • Usability scores (System Usability Scale = a standard 10-question survey for usability)
Qualitative insights	• Semi-structured interviews (guided conversations to know the user experience) • Think-aloud protocol (users verbalize what they think while using the system)

Table 1 Evaluation Measures and Methods of DDW in Beta Testing

Note: *Quantitative metrics are used to evaluate task functional efficiency, accuracy, and system-wide usability., while qualitative insights aim to understand participants' user experience, perceptions regarding trust, data sovereignty, and satisfaction with the reward system in the DDW. Qualitative insights are prioritized in this phase to explore the nuances of user trust in decentralized fintech architectures, which are often more critical for adoption than raw speed.*

2.3 Discussion on Limitations & Practicality: Empirical Mixed-Methods Analysis

This section adopts an empirical mixed-methods approach to assess the real-world practicality and deployment readiness of the Decentralized Data Wallet (DDW). The analysis triangulates quantitative system-level evidence with qualitative assessments of industry feasibility, enabling a more comprehensive evaluation than either method alone.

On the quantitative side, the evaluation examines objective performance indicators, including the ***accuracy of derived data insights*** (regarding the ML performance) and ***tokenomics efficiency*** (regarding the Blockchain Performance), in addition to observable system overheads introduced by on-chain operations. These metrics provide empirical evidence of the protocol's technical capabilities and operational constraints.

A qualitative investigation complements the empirical analysis through semi-structured interviews with industry practitioners, including credit analysts and fintech professionals. These interviews provide valuable expert insight into the non-technical factors that significantly influence the adoption of the DDW, such as ***challenges in institutional integration***, ***regulatory compliance requirements***, and ***scalability limitations*** inherent in current blockchain infrastructures. Practitioner feedback is used to contextualize the empirical findings, ensuring that they align with existing financial workflows and governance frameworks.

The discussion focuses the perceived effectiveness of insight accuracy and tokenomics efficiency, while also addressing broader institutional challenges, particularly those related to regulatory compliance and scalability. By prioritizing practitioner perspectives, this evaluation identifies both the strategic limitations and the readiness of the protocol for industrial deployment, offering a comprehensive view of its practical viability within the current or future decentralised financial landscape.

3. Current Progress

This section presents the current progress and our development status of the DDW prototype development lifecycle. Section 3.1 begins with the findings from the MLR, which establishes the foundational requirements that inform the design and architecture of the system. Section 3.2 details the system architecture and protocol design of the DDW. Section 3.3 focuses on the user journeys for both data contributors and consumers, mapping each interaction flows and identifying key touchpoints.

Section 3.4 shows the current frontend implementations of the DDW. Section 3.5 detailed the explanations on the backend implementations, including the backend operational logic of the DDW and the backend ML algorithms for DPI generations. Section 3.6 discusses the API interface, which bridges the off-chain analytics engine with the frontend system, enabling seamless interaction between components. Section 3.7 examines the technical limitations encountered during the development and the corresponding mitigations employed. Finally, Section 3.8 summarizes the current project status and outlines the future work plan.

3.1 Findings from MLR

The purpose of this MLR is to synthesize insights from both academic research and grey literature to establish the foundational requirements for the DDW in our project. This review systematically evaluates existing blockchain-based data infrastructures while gaining insights from current industry trends to inform the project's system architecture, protocol design, operational workflow and prototype.

3.1.1 Comparative Analysis of Blockchain Data Infrastructures

Table 2 compares the four primary paradigms and industry applications identified during the MLR. These cases represent a broad architectural spectrum, ranging from identity management to autonomous insight generation.

Case Study	Type	Spectrum Representation	What is it doing?	Core Problem Solved	Novelty / Key Contribution	Key Standards / Technical Mechanisms
Civic (Civic Pass) [19], [20]	Industry	Identity & Authentication	Issuing "passes" (represented by Soulbound Tokens (SBTs)) to verify user identity directly on-chain.	Eliminates repetitive Know Your Customer and prevents Sybil attacks in P2P markets.	Links real-world identity to wallet addresses using SBTs [19].	W3C DIDs [20], SBTs [19]
DataMesh+ [6]	Research	Data Infrastructure & Exchange	Managing P2P data exchanges through decentralized access control.	Resolves bottlenecks in centralized IoT data marketplaces.	A hybrid P2P exchange model for high-performance self-sovereign data [6].	Decentralized Access Control (DAC), P2P Protocols [6]
Ocean Protocol [7], [17]	Industry	Data Monetization & Assetization	Converting data access rights into liquid financial assets (Data Tokens).	Breaks down data silos by creating a standardized monetization layer.	Tokenizes access rights to create a liquid, privacy-preserving data economy [7].	ERC-721 (Data NFTs), ERC-20 (Datatokens) [6]
Insight Genesis [18]	Industry	Artificial Intelligence (AI) & ML Application	Transforming raw datasets into verifiable, actionable insights via AI.	Solves the "data noise" problem where raw data is too complex for consumers.	Focuses on the output (insights) rather than the input, streamlining ingestion [18].	Verifiable Computation, AI Oracle integration [18]

Table 2 Comparative Analysis of Foundational Blockchain-based Data Infrastructures and Paradigms

Note: *This table provides a multidimensional comparison of the primary research and industry frameworks evaluated during the MLR. Essential components are identified for building the DDW's decentralized data-to-insight pipeline for the DDW prototype.*

3.1.2 Cross-Synthesis of MLR Findings across Academic and Industry Literature

The findings from the Multivocal Literature Review (MLR) reveal a fundamental architectural evolution within the decentralized data landscape: a shift from rudimentary peer-to-peer storage models toward intelligent, verifiable data ecosystems that prioritize utility alongside sovereignty. A primary finding, synthesized from the peer-reviewed work of Merlec and In [6] regarding DataMesh+ and the industry-standard protocols advocated by McConaghy [17] of Ocean Protocol, is the operational necessity of decoupling identity from raw behavioral data. This is achieved through the implementation of DIDs and SBTs, as pioneered in the "Self-Sovereign Identity" (SSI) manifesto by Christopher Allen [19] and formalized by the W3C standards [20]. These frameworks facilitate a "trustless" verification layer where the user retains absolute ownership of their digital footprint while providing granular, permissioned access to third parties for specific computational tasks.

Furthermore, the synthesis identifies a critical "utility gap" in current frameworks: while robust infrastructure exists for secure data hosting, there remains a significant deficit in systems capable of autonomously transforming high-entropy, raw personal data, into structured predictive metrics or verifiable ML insights without compromising underlying privacy. The documentation for Insight Genesis [18] addresses this by shifting the focus from simple data assetization to verifiable insight generation. This conceptual shift directly informs the DDW's internal logic, enabling a seamless pipeline that converts user-held data into high-value Machine Learning (ML) insights.

3.1.3 Contribution to DDW Logic and Concept Formation

Collectively, these findings mandate the DDW's adoption of a hybrid authentication model and an evolving machine learning architecture. This research directly informs the DDW's **"Contribute-to-Earn"** economic logic, utilizing the **Web3 Data Value Chain [17]** to transform the user from a passive data subject into an active participant. This mechanism ensures a sustainable equilibrium between data supply and computational demand, mirroring the **"Compute-to-Data"** staking models observed in **Ocean Protocol [6]**.

To realize this, the our DDW's model partitions the process into two distinct stages:

- **A Local Privacy layer:** Where raw personal behavioral data remains within the user's control, stored via the encrypted InterPlanetary File System (IPFS), and is processed off-chain to ensure that sensitive personal information never enters the public ledger or a centralized database.

- **An On-chain Verification Layer:** Where only the derived insights, strictly excluding raw data, are validated, finalized, and minted as Data NFTs (Access Licenses) for direct consumption by data consumers.

The execution of 2-layered model is facilitated by a **three-token system**, which codifies the relationship between data ownership, insight utility, and tokenomics framework within the DDW protocol (see Table 3).

Token Type	Category	Purpose	Transferability
Data Ownership Token	**Soulbound Token (SBT)**	Acts as a "Digital Deed" proving the user is the verified owner of the raw data.	**Non-Transferable**
Insight Token (DPI Token)	**Data NFT/License**	A cryptographic summary of a specific result that acts as the "Access Ticket" for a consumer.	**Access-Controlled**
Reward Token	**Utility Token**	The fungible currency used by enterprises to pay for access and by users to receive incentives.	**Transferable & Redeemable to ETH**

Table 3 Tokenomics Framework in DDW Application

Note: *This table summarizes the functional roles, categories, and transferability constraints of each token type within the DDW ecosystem, illustrating how the protocol enforces data sovereignty while supporting controlled data monetization.*

Figure 2 illustrates how these three token types interact and flow between data contributors and consumers within the DDW ecosystem.

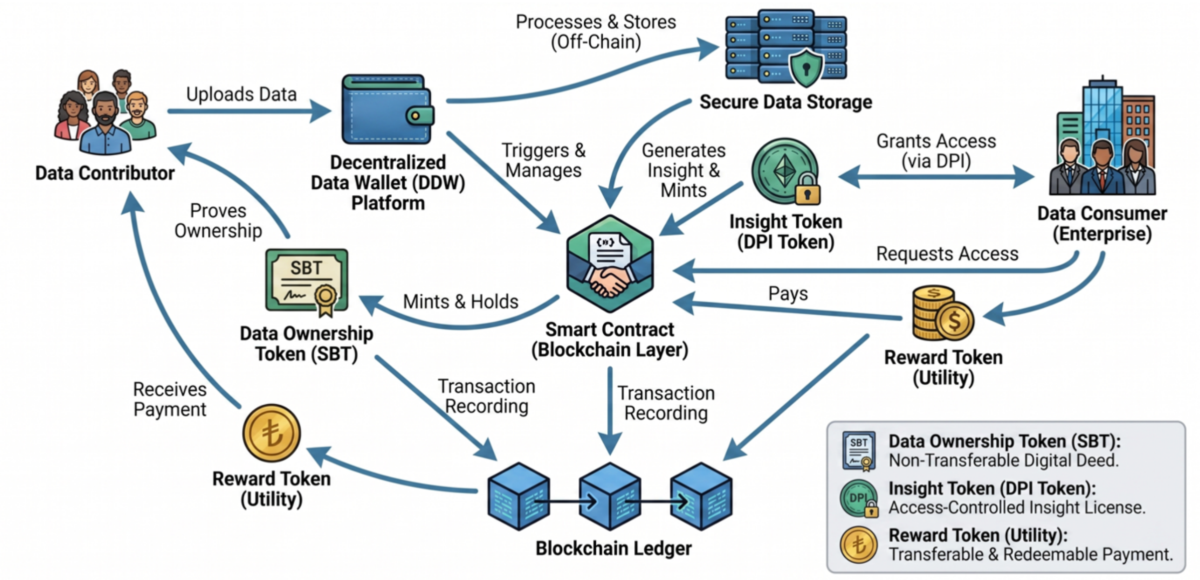

Figure 2 Interaction and Flow of Tokens Between Data Contributors and Consumers in the DDW Tokenomics Framework

The DDW prototype was initially developed as a dual-layered hybrid ML architecture, combining on-chain identity verification and tokenization with off-chain model execution. While future iterations aim for full on-chain ML execution in the upcoming semester, where the ML algorithm runs directly within smart contracts, the current design ensures that raw data remains private off-chain, while the blockchain anchors and verifies the resulting insights, maintaining transparency, immutability, and trust.

Thus, in the following sections, all descriptions of system architecture, workflow, and implementations are based on this hybrid off-chain ML design.

3.2 Workflow, System Architecture and Protocol design of DDW

The end-to-end operational progression of the Decentralized Digital Wallet (DDW) is illustrated in **Figure 3**. This workflow is designed to map the sequential interactions between the user interface and the decentralized backend, ensuring that each data transaction is verified through the hybrid off-chain ML model before state changes are committed to the blockchain.

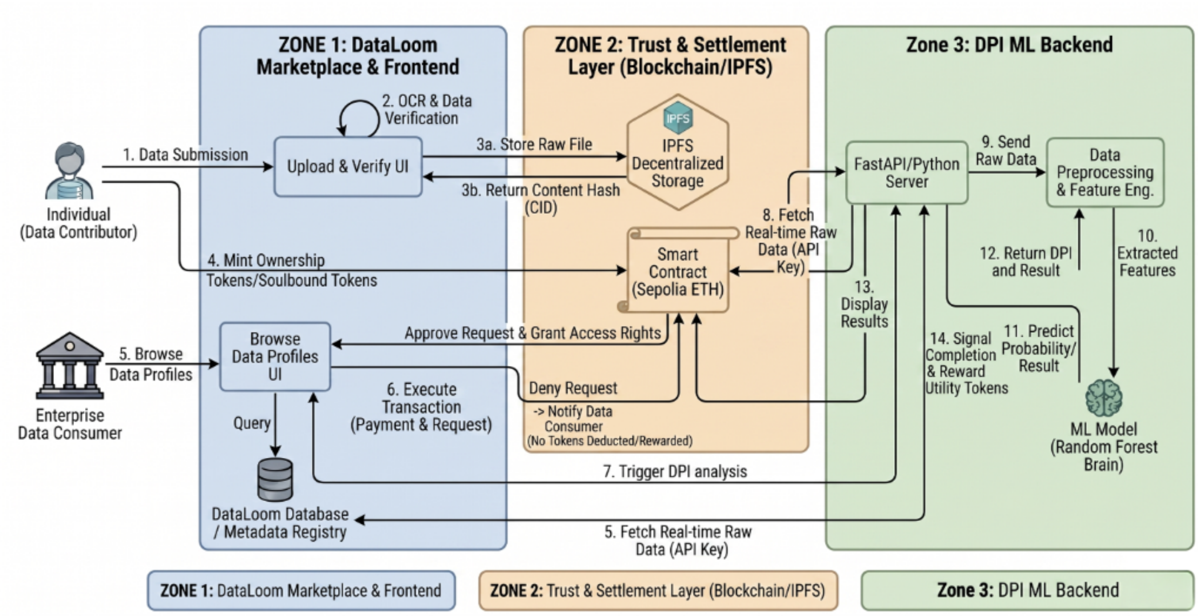

Figure 3 End-to-End Workflow of DDW

The DDW system architecture is organized into a four-tier modular protocol stack comprising the Base, Core, Service, and Application Layers, complemented by a set of supporting components (see Figure 4 & Table 4).

The ***Base Layer*** provides the foundational infrastructure of the DDW, leveraging the Ethereum blockchain and the InterPlanetary File System (IPFS). Ethereum serves as an immutable, tamper-proof ledger for transaction and token records, while IPFS offers low-cost, decentralized storage for encrypted raw travel data. This hybrid storage approach offloads high-entropy data from the blockchain to enhance scalability and cost-efficiency, while maintaining a verifiable link to the original data via on-chain Content Identifiers (CIDs).

The ***Core Layer*** defines the operational logic and trust framework of the protocol, consisting of smart contracts and Ethereum's Proof-of-Stake (PoS) consensus mechanism. They together enable trustless verification without relying on a central authority. These smart contracts are self-executing programs that automatically enforce the rules of the DDW, such as verifying data submissions, issuing ownership tokens, and managing granular access permissions. By automating these processes on-chain, the Core Layer ensures that the DDW's operations remain consistent, transparent, and resistant to unauthorized tampering.

The ***Service Layer*** acts as the functional engine that transforms verified data into tokenized digital assets. Within this layer, a decentralized data verification process confirms the authenticity of uploaded records, while the tokenization engine mints non-transferable Data Ownership Tokens (SBT) to bind data permanently to the contributor's wallet. This layer also houses the Insight Generation ML Engine, which utilizes a machine-learning model to produce anonymized insights, such as the alternative credit scores demonstrated in this prototype. The Service Layer incorporates user access and control mechanisms that allow data contributors to review, approve, or reject data consumer requests from the data consumers. Users can also define precise and time-bound permissions enforced directly by smart contracts in this function. Thus, these components in the Service Layer collectively support the core function of DDW: turning verified data into tokenised digital assets.

The ***Application Layer*** provides the user-facing interfaces for interaction between the two primary stakeholders: Data Contributors and Data Consumers. It consists of two primary frontend components. The Data Contributor interface, implemented as a decentralized user wallet, enables individuals to manage their data sovereignty by securely uploading personal data, controlling consent and access permissions, managing non-transferable ownership and identity tokens, and tracking protocol-based rewards. In parallel, the Data Consumer interface is delivered as an enterprise dashboard which allows them to browse and view the "personal data profiles" of various data contributors and initiate smart-contract-governed requests to unlock and access specific insights. By settling the payment through the on-chain transfer of DDW platform's utility tokens data consumers obtain permissioned access to view the insights upon successful approval.

These four-layered architectural stack is the augmented by a suite of Supporting Components that manage the protocol's economic and security resilience. The Privacy-Preserving Protocols ensure data anonymity during the computation cycle, while the Utility Tokenomics Framework programmatically manages the lifecycle of the DDW's utility token. This framework governs the minting, distribution, and staking logic, ensuring that rewards are disbursed only upon the successful cryptographic "unlocking" of insights. Furthermore, the protocol prioritizes Security and Auditing through Smart Contract Verification, utilizing transparent, audited code to maintain a trustless environment.

As for other system properties related to performance, the DDW demonstrates strong scalability through its multi-layered offloading strategy. First, by decoupling computationally intensive machine learning tasks and high-volume data storage (via IPFS) from the main blockchain, the architecture avoids throughput bottlenecks and excessive gas costs typically associated with monolithic on-chain designs. Second, to further enhance transaction efficiency, the protocol is designed for deployment on an Ethereum Layer-2 (L2) scaling solution that leverages rollup technology to batch transactions off-mainnet while preserving Layer-1 security guarantees. Although the current prototype is validated on the Ethereum Sepolia testnet, the EVM-compatible architecture ensures that high-frequency minting and reward distribution cycles remain economically viable as user participation scales. This modular design also provides a resilient foundation for the planned transition toward full on-chain ML execution in the coming semester, where the expanded block space offered by an L2 environment will be essential for trustless computational workloads.

Overall, the proposed four-tier architecture achieves a rigorous separation of concerns across foundational infrastructure, execution logic, and stakeholder interfaces. It should bt noted that refinements would be needed to verify the effectiveness of the DDW in real-world deployment.

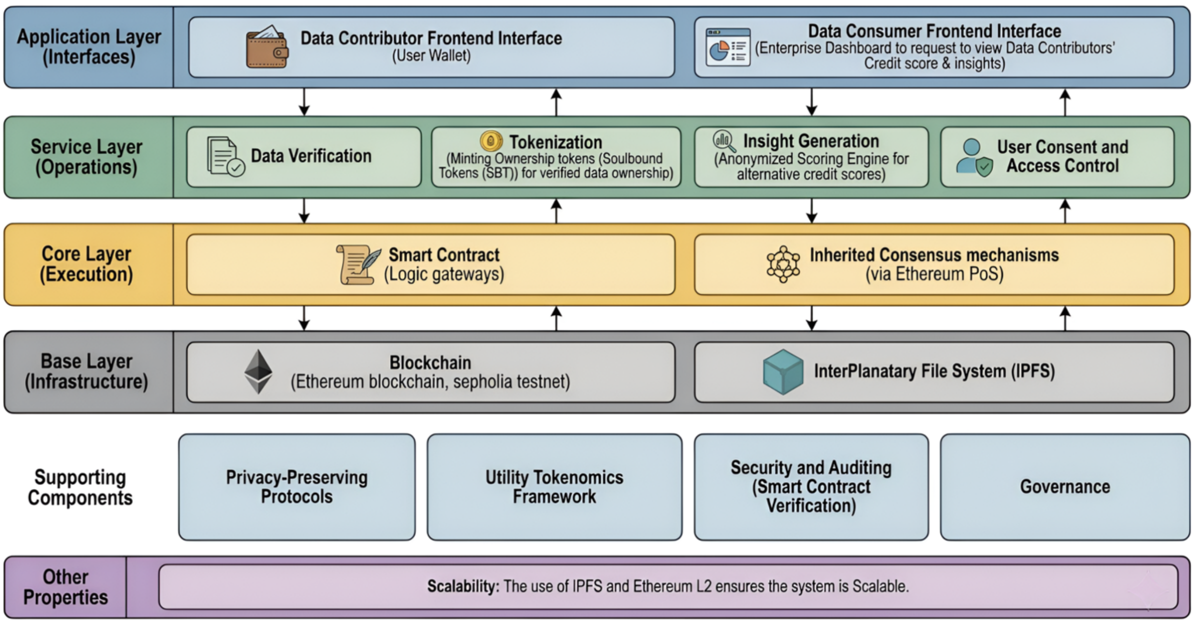

Figure 4 The Four-Layered System Architecture Overview of DDW

Layer	Components
Base Layer (Infrastructure)	• Blockchain (Ethereum blockchain, Sepolia testnet) • InterPlanetary File System (IPFS)
Core Layer (Execution)	• Smart Contract (Logic gateways) • Inherited Consensus mechanisms (via Ethereum PoS)
Application Layer (Interfaces)	• Data Contributor Frontend Interface (User Wallet) • Data Consumer Frontend Interface (Dashboard to request to view all the data contributors' data profile & request to unlock and access to one's Insights)
Service Layer (Operations)	• Data Verification • Tokenization (Minting Ownership tokens for verified data ownership) • Insight Generation (Anonymized Scoring Engine for • User Consent and Access Control
Supporting Components	• Privacy-Persevering Protocols • Utility Tokenomics Framework (Minting utility tokens & Handling staking tokens for rewards and conversions to ETH) • Security and Auditing (Smart Contract Verification) • Governance
Other properties	• Scalability

Table 4 DDW System Architecture Layers and Components

***Note**: This table summarizes the system architectural layers of the DDW and the key components in each layer to ensures functional clarity and operational coherence.*

3.3 User Journey Mapping

Section 3.3 describes the user journeys of the DDW prototype from the perspectives of its two primary stakeholder groups. It documents the sequential flow through which each parties (Data contributors, as individuals & Data consumers, as enterprises) progress within the system, from entry to task completion.

3.3.1 The Data Contributor Journey

The data contributor's journey is built on the principle of data sovereignty, ensuring that every action, from data submission to reward, remains transparent and under the user's absolute control.

In **Stage 1, the Multi-Layered Onboarding and Identity Setup stage**, the user enters the DDW application and begins by authenticating through a Social SSO (e.g., Google, Microsoft, or Facebook). This initial layer provides a familiar, low-friction entry point for non-crypto users. Following successful social authentication, the user is prompted to connect their MetaMask wallet. This step maps the user's social identity to a unique cryptographic address on the Sepolia Testnet, establishing a secure decentralized identity that serves as the permanent anchor for their data assets and transaction signatures.

In **Stage 2, the Data Submission and Secure Upload stage**, the user selects specific behavioral logs to share. During the secure upload process, the data is encrypted and transferred to the private off-chain repository. The interface provides immediate feedback, reassuring the user that raw data remains private and is never exposed on the public ledger.

In **Stage 3, the Ownership Verification (SBT Minting) stage**, following a successful upload and quality check, the user's dashboard updates to display a new SBT. Acting as a "Digital Deed," this token provides visual confirmation that ownership of the specific dataset has been officially anchored to the blockchain.

In **Stage 4, the Consent and Access Management stage**, the system generates insights (e.g., an alternative credit score), triggering real-time notifications for enterprise access requests. The user reviews the request details—including the requester's identity and offered reward—and chooses to Approve or Deny. No data exchange occurs without this explicit cryptographic consent.

In **Stage 5, the Reward Distribution and Asset Tracking stage**, upon approval and successful access by the enterprise, the system triggers an automated payment. The user receives Utility Tokens directly in their wallet. The journey concludes at the Rewards Dashboard, where users monitor historical earnings, identify which insights are generating value, and track the overall valuation of their data portfolio.

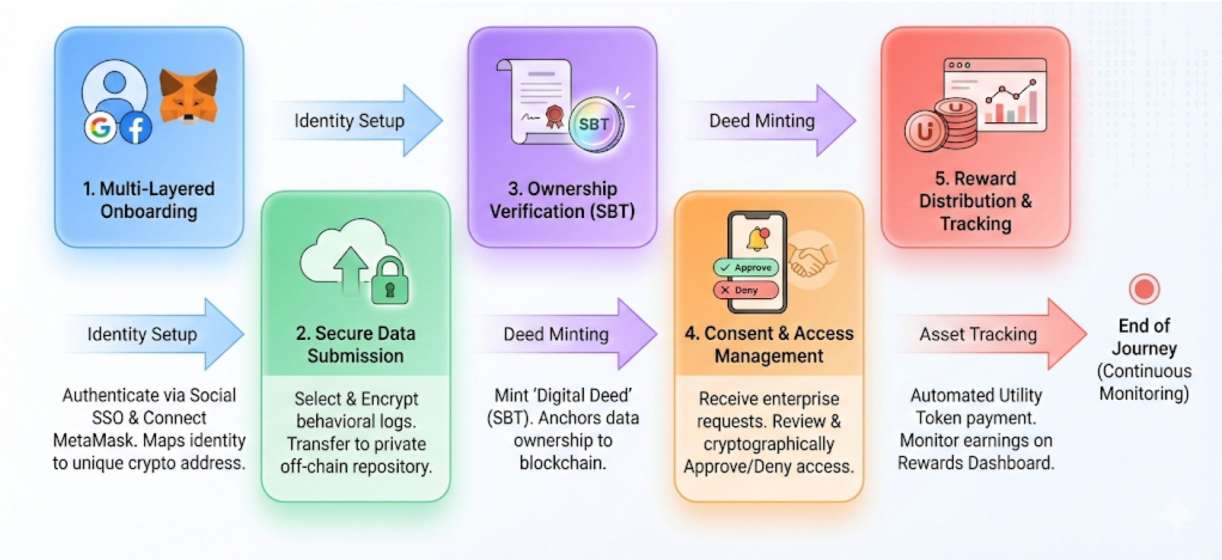

Figure 5 Data Contributor User Journey Flow in the DDW

3.3.2 The Enterprise Data Consumer Journey

The enterprises consumer's journey is optimized for Efficiency and Trust, enabling businesses to acquire high-fidelity, verified insights without the legal and security liabilities associated with handling raw personal data.

In **Stage 1, the Enterprise Onboarding and Identity Selection stage**, the enterprise begins by authenticating through a specialized corporate login (e.g., Google or Microsoft Business SSO) to establish their professional profile. Following this, the enterprise connects an existing institutional wallet (such as a corporate MetaMask or Gnosis Safe Multi-Sig) to link their pre-funded treasury to the platform. This two-tier process establishes the enterprise's cryptographic identity within the DDW ecosystem, ensuring they possess the necessary Utility Tokens to engage with the decentralized protocol and initiate access requests for high-value generated insights.

In **Stage 2, the Data Discovery stage**, the enterprise logs into a specialized dashboard to browse anonymized "Data Profiles." Rather than viewing personal identities, they filter contributors based on behavioral criteria (e.g., "Frequent Travelers" or "Consistent Commuters") to identify segments relevant to their strategic needs.

In **Stage 3, the Access Request and Escrow stage**, after identifying desired insights, the enterprise submits a formal access request. This process involves committing Utility Tokens to a smart contract escrow, which serves as a financial guarantee of payment once the contributor grants permission.

In **Stage 4, the Insight Consumption and Strategic Use stage**, once the contributor approves the request, the enterprise "unlocks" the DPI Token. The verified result (e.g., the finalized credit score) becomes visible through their dashboard, accompanied by cryptographic proof of authenticity. This ensures the insight is derived from real, verified behavior without the enterprise ever accessing the underlying raw data logs.

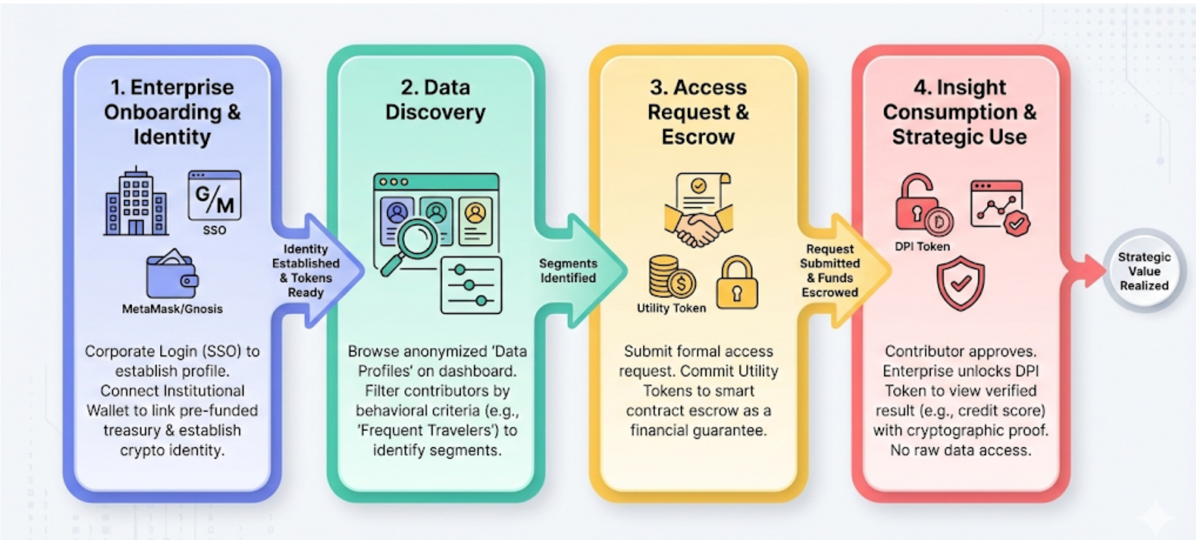

Figure 6 Data Consumer User Journey Flow in the DDW

By mapping these distinct yet interconnected journeys, the DDW transforms the traditionally opaque process of data collection into a transparent, bilateral exchange of value. Ultimately, this user-centric design ensures that while the underlying blockchain operations maintain technical rigor and trust, while the end-user experience remains intuitive, fostering a sustainable ecosystem for decentralized data sovereignty.

Sections 3.4 and 3.5 present the frontend and backend implementations of the DDW. The current implementation is preliminary and is designed to support a single data domain and application scenario for foundational development and demonstration purposes. Specifically, the prototype focuses on travel behavioral data used to generate insights related to alternative credit scoring, as mentioned in Section 1.4. The following sections outline the basic user journey and data flows to support conceptual clarity in this interim report, as illustrated in the specific use-case workflow below (see Figure 7). Further refinements and extensions to include additional data domains will be undertaken in the following semester.

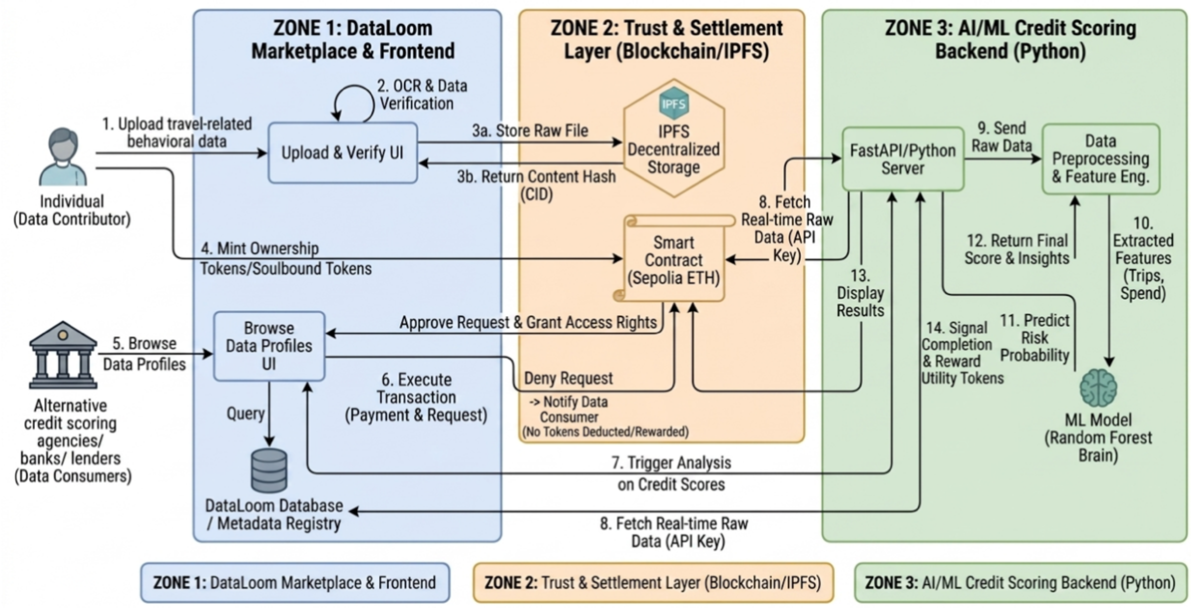

Figure 7 Specific Use Case Demonstration: End-to-End Workflow of DDW for Insight Generations on Alternative Credit Scores & Relevant Credit Insights using Travel-Related Behavioral Data

3.4 Frontend Implementations of the DDW

Section 3.4 presents the current frontend implementation of the DDW, detailing the interfaces for data consumers (Section 3.4.1) and data contributors (Section 3.4.2). It outlines the core features currently integrated into the system and their respective functions. The DDW frontend is implemented as a decentralized application (dApp) designed to support the secure transfer of data value. The subsections that follow describe the current interface state and the features that enable the "Data-to-Asset" lifecycle.

3.4.1 Data Contributor Interface: Data Ingestion and Tokens Management

This section presents the functional interfaces developed for individual data producers, highlighting the transition from traditional authentication to decentralized asset control.

To minimize entry barriers for non-technical users, the DDW employs a Unified Login Interface that provides a standardized entry point for both data contributors and consumers. Users begin the onboarding process by authenticating through a Social SSO provider, such as Gmail, Microsoft, or Facebook—which ensures the first touchpoint is intuitive and avoids the immediate friction of blockchain technicalities (see Figure 8).

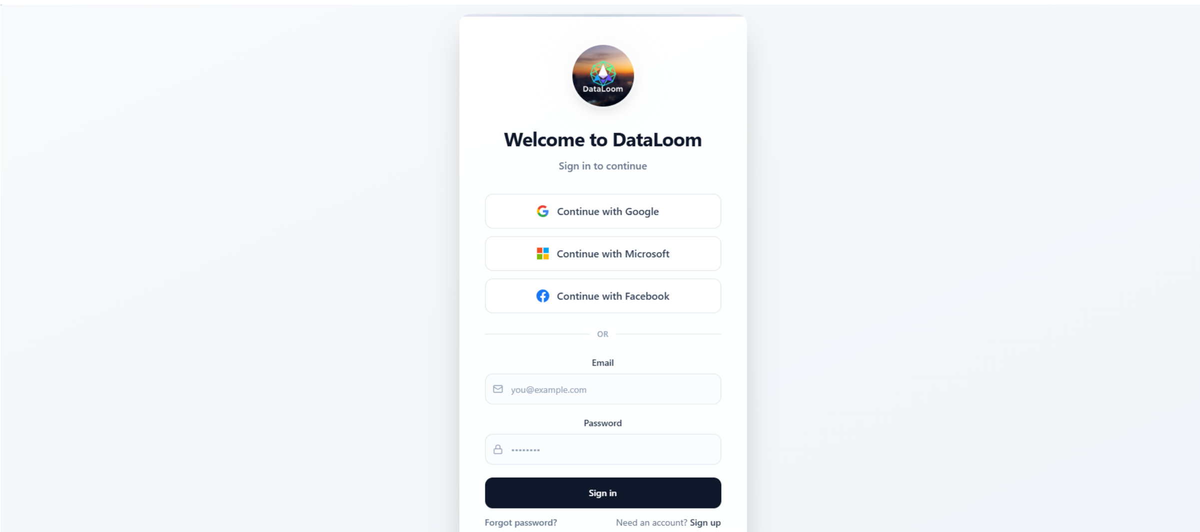

Figure 8 Unified Onboarding Interface

Upon successful authentication, the system redirects users to the DataLoom Homepage, which currently serves as the central hub for all data management tasks (see Figure 9). While the current prototype displays separate designations for "Data Producers" and "Data Consumers," these labels are slated for further consolidation in upcoming iterative refinements to create a truly seamless, role-agnostic landing environment.

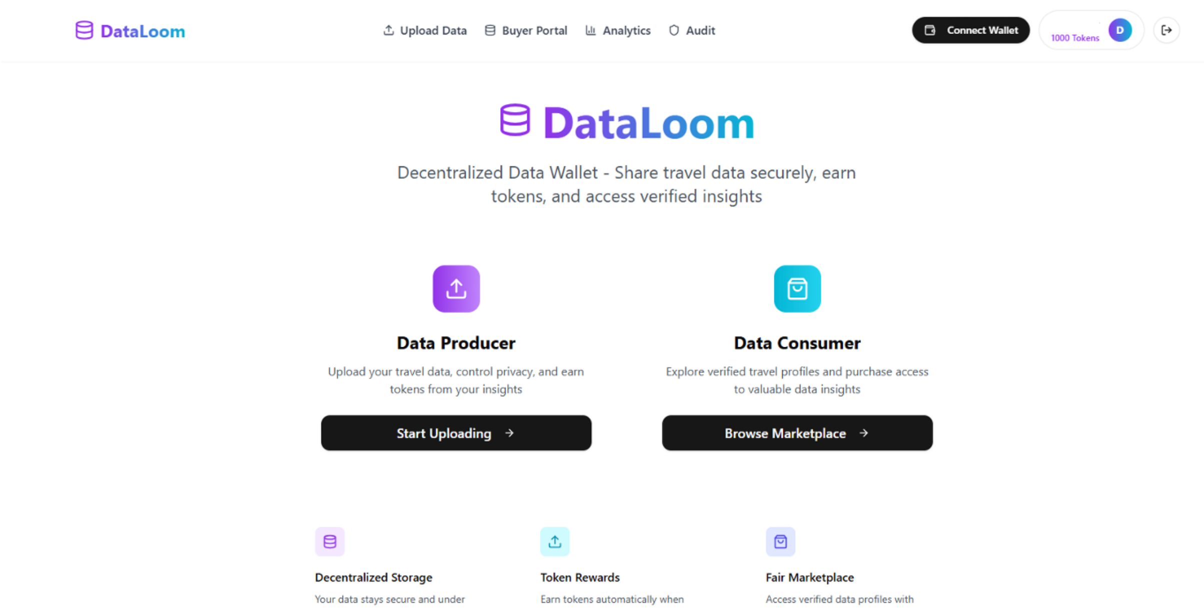

Figure 9 DataLoom – DDW's Homepage

Since the wallet connection serves as the definitive login credential for the decentralized layer, the interface removes the need for traditional password management by utilizing a direct cryptographic handshake. Users are prompted to locate the "Connect Wallet" button positioned in the top-right corner of the navigation bar (see Figure 10).

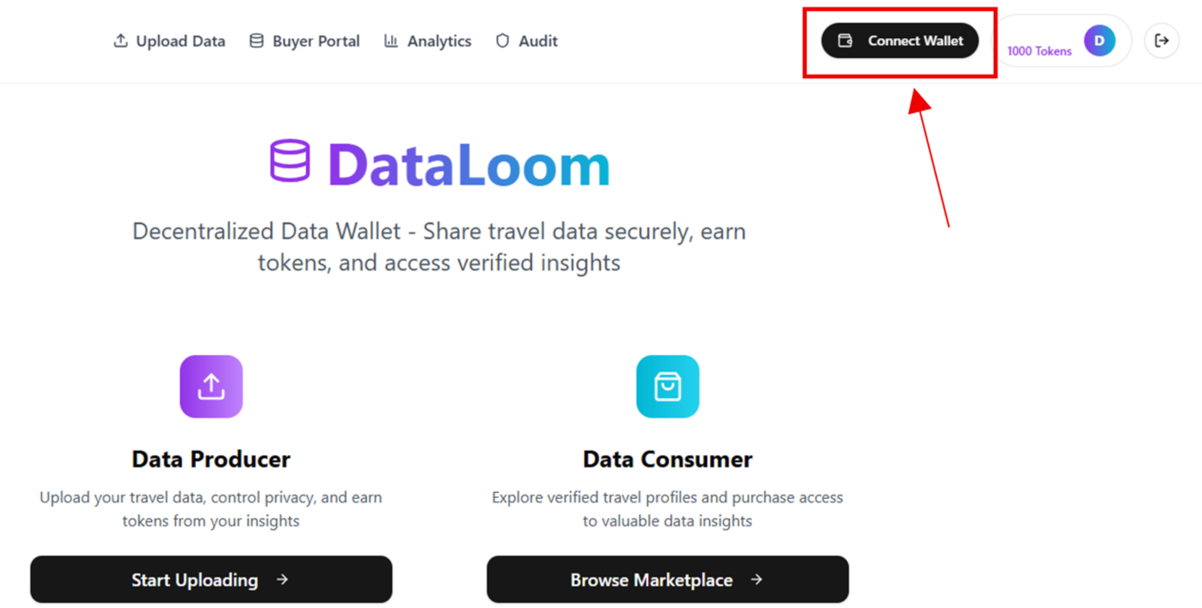

Figure 10 Primary Wallet Connection Interface

Upon clicking this button, a MetaMask pop-up is triggered, requiring the user to approve the connection request so that DataLoom can recognize their unique address. To ensure compatibility with the prototype's backend, users must verify that their wallet is set to the correct network, specifically the Sepolia Testnet. Once the transaction is signed and the connection is established, the button dynamically refreshes to display a shortened version of the user's public wallet address (see Figure 11), signaling that the session is now live and fully authenticated on the blockchain.

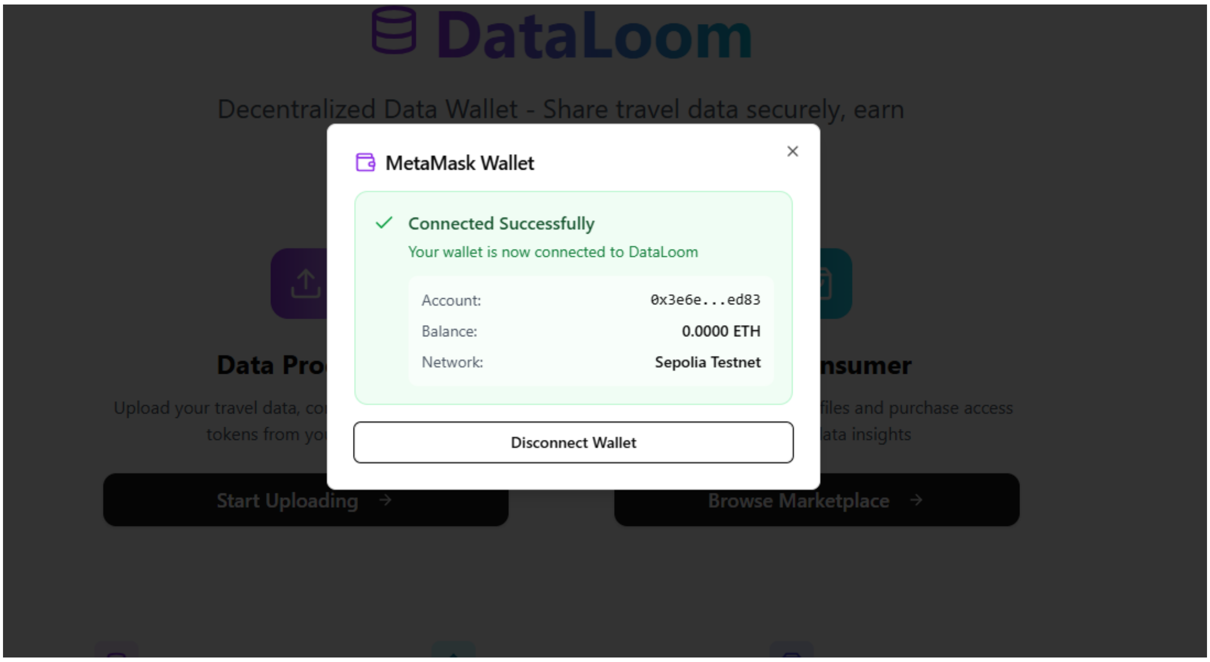

Figure 11 Authenticated Status Upon Successful Connections to MetaMask Wallet

Note: *This figure displays the finalized connection state between DDW and data contributor's MetaMask wallet, revealing the contributor's wallet public address hash, active operating network (Sepolia Testnet), and current ETH balance for transaction transparency*.

Next, during the data uploading and minting stage, raw behavioural data files are converted into blockchain-anchored digital assets through localized encryption and automated valuation processes.

Once the identity is established, contributors navigate to the upload interface by selecting the "Start Uploading" call-to-action on the primary dashboard. The interface presents a structured ingestion form where users input essential metadata, including a Trip Name, Category, and Description, to define the context of their dataset (see Figure 12). After populating these fields, the user selects a data file (supporting JSON, CSV, or PDF formats) via a dedicated drag-and-drop zone. At this stage, the frontend executes a local encryption protocol to ensure data privacy before the file is transmitted to the Distributed Storage Layer (IPFS).

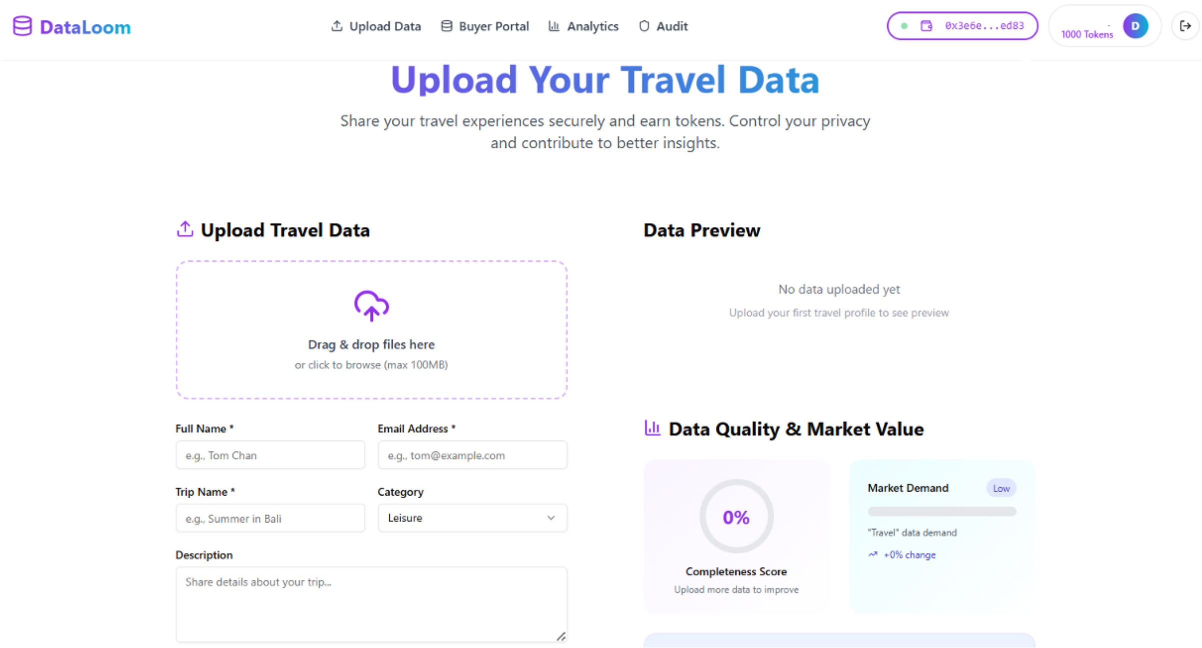

Figure 12 Data Ingestion and Metadata Interface before Uploading Data

Upon clicking the "Upload and Analyse" button, the system initiates a real-time backend evaluation of the submission. The interface dynamically updates to present a Data Preview and a comprehensive Data Quality & Market Value assessment (see Figure 13). This implementation specifically highlights three integrated features:

- **Data Quality (Completeness Score):** An automated validation protocol that calculates a percentage-based score (e.g., 76%) to determine if the uploaded data is readable, standardized, and complete.

- **Market Demand Assessment:** A dynamic indicator that matches the user's specific data type against current buyer requests in the marketplace to estimate its desirability.

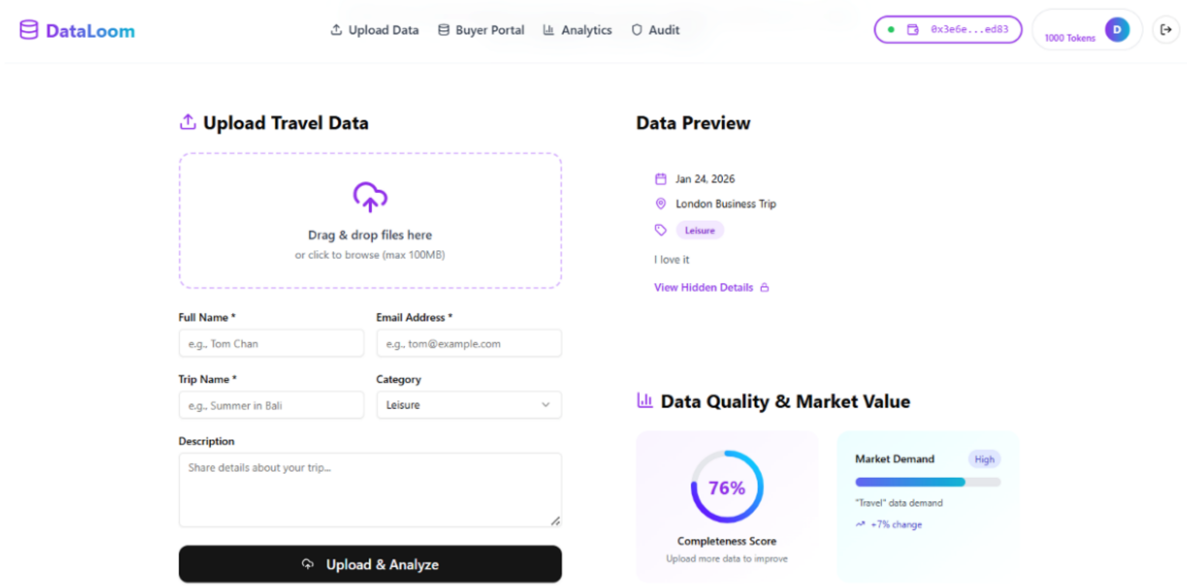

Figure 13 Updates on Data Quality Score and Real-Time Market Demand Trends after Uploading Data

Notes*: The Data Quality Score measures structural completeness and readability, while the Market Demand Trends match the data against enterprise request volume.*

Following the successful analysis, the system calculates and displays the Token Reward Allocation, which is a specific amount of platform tokens credited to the user's internal balance as an immediate incentive based on the data's quality and current market demand. Contributors can then manage the long-term governance of these assets through the Privacy & Permissions and Token Reward dashboard (see Figure 14). This integrated interface provides granular control, allowing users to toggle anonymization settings and authorize enterprise access requests.

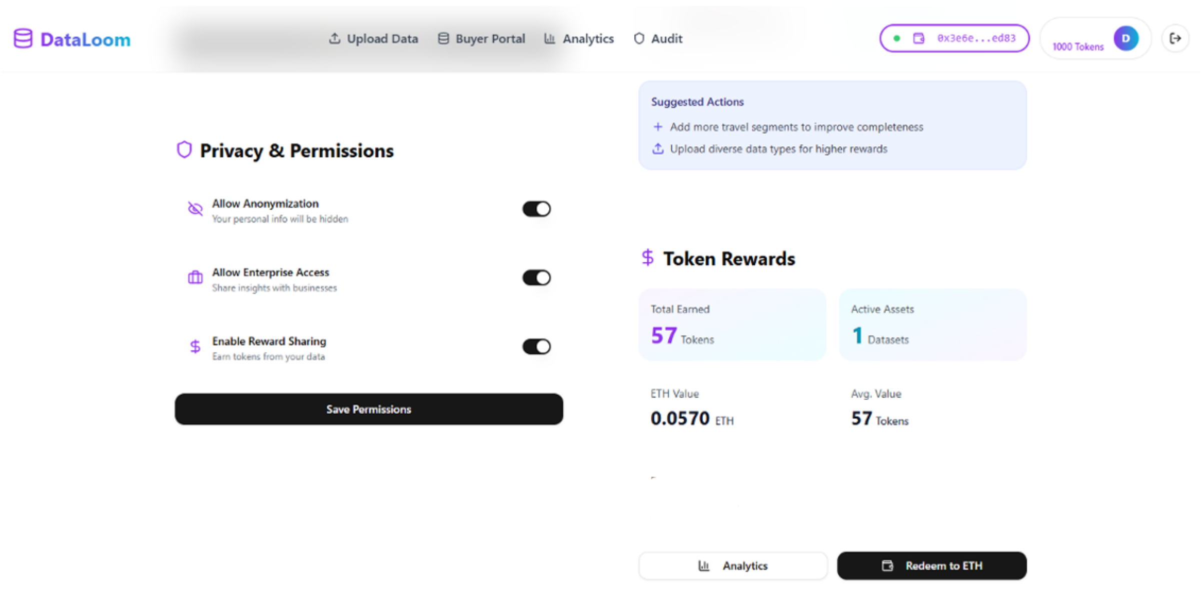

Figure 14 Privacy Governance and Token Rewards Dashboard

The Token Rewards Dashboard serves as the primary interface for the financial closure of the "Data-to-Asset" cycle, providing contributors with the necessary tools to convert platform incentives into liquid ETH assets.

The Token Rewards Dashboard serves as the primary interface for completing the “Data-to-Asset” cycle, enabling contributors to convert earned platform incentives into liquid ETH. It provides a transparent overview of cumulative rewards, which are distributed only when Data Consumers purchase access to a contributor’s DPI, ensuring that compensation reflects the market value and demand of the data. As shown in Figure 14, the dashboard displays the total utility tokens alongside their real-time ETH valuation, allowing users to track the liquidity and market power of their earnings. Central to this module is the “Redeem to ETH” feature, which securely initiates the cryptographic process to convert protocol rewards into liquid ETH.

To ensure the security of the value exchange, Figure 15 provides a Magnified View of Token Rewards. This granular close-up highlights the specific network metadata and connection status required for the transaction. This step ensures that users can confirm their non-custodial MetaMask wallet is properly connected to the Sepolia Testnet before authorizing the smart contract to execute the automated conversion of their earned DDW platform’s utility tokens to ETH.

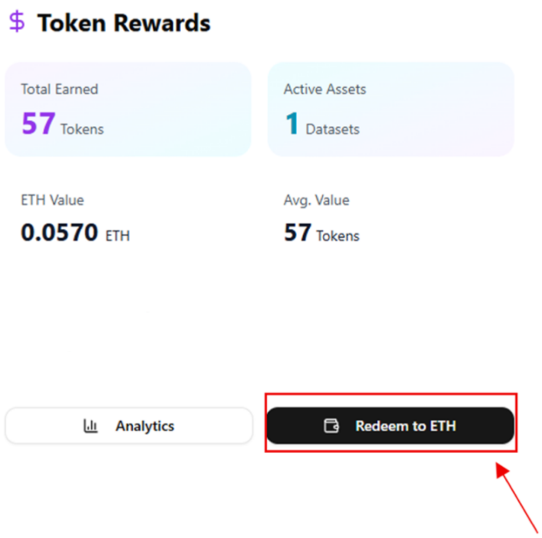

Figure 15 A Close-up View of the Token Rewards Dashboard

After selecting “Redeem to ETH” in Figure 15, the Redeem Rewards interface pops up. . To allow indivudals to convert their DDW utility tokens into ETH, users must first click the “Connect Wallet” button (see Figure 16), to establish the financial bridge between DDW and their MetaMask Wallet.

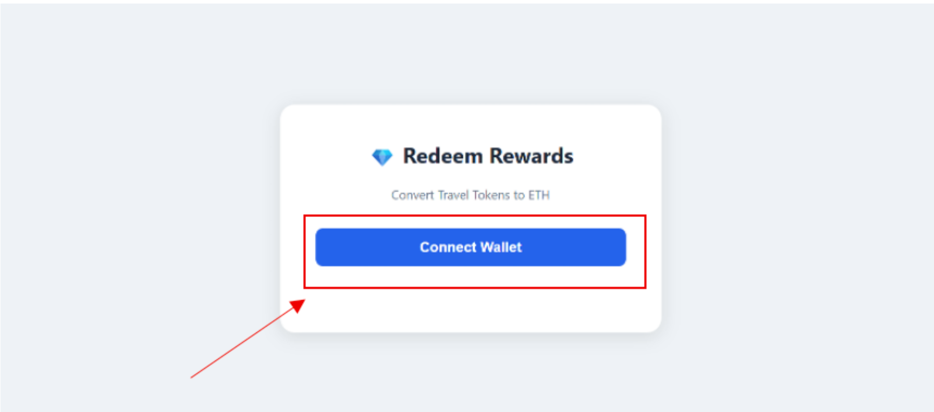

Figure 16 Reward Redemption and Wallet Authorization Gateway

Once the connection is live, the user is prompted to type the specific amount of ETH they wish to transfer to their MetaMask wallet based on their earned token balance. The process concludes when the user clicks the green "Confirm Swap" button (see Figure 17), which triggers the smart contract to automatically transfer the earned utility tokens the DDW platform directly to the user's verified wallet address.

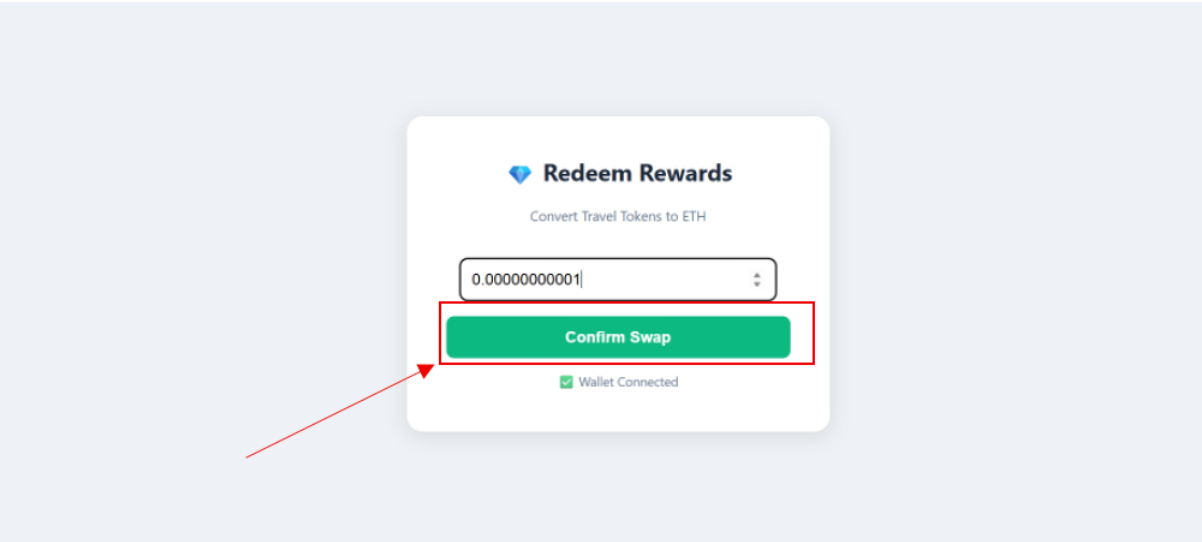

Figure 17 Reward Liquidation and Atomic Swap Finalization Interface

To finalize the Value-Exchange Cycle of token redemption and asset liquidity, the platform provides a robust cryptographic verification feature, ensuring a secure and transparent transition from internal protocol incentives to liquid external assets. This functional layer utilizes a dual-transaction framework to maintain the integrity of the platform's liquidity pool and prevent unauthorized asset movement.

Before any on-chain interaction is executed, the system enforces a final security checkpoint. As shown in Figure 18, the MetaMask Login Interface requires users to authenticate by entering their wallet password, decrypting the non-custodial wallet and granting access to the private keys necessary to authorize the smart contract. Only after this explicit confirmation can the redemption of DDW utility tokens to ETH proceed, ensuring that all value transfers occur with full user consent and cryptographic authorization.

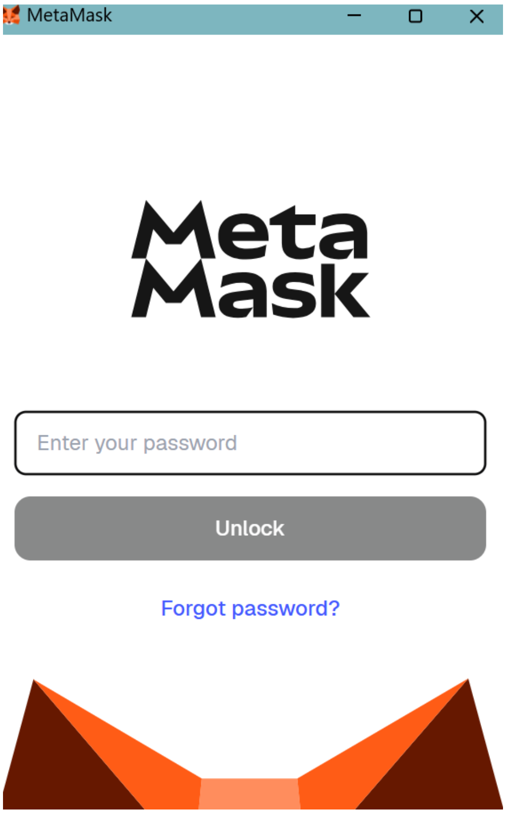

Figure 18 MetaMask Wallet Login and Secure Authentication Interface

Once authenticated, the platform invokes the MetaMask Spending Cap Request feature and the user should click “Confirm” to continue the ETH redemption (see Figure 19). This is a critical ERC-20 security standard that grants the smart contract explicit permission to interact with the contributor's specific token balance on the Sepolia Testnet.

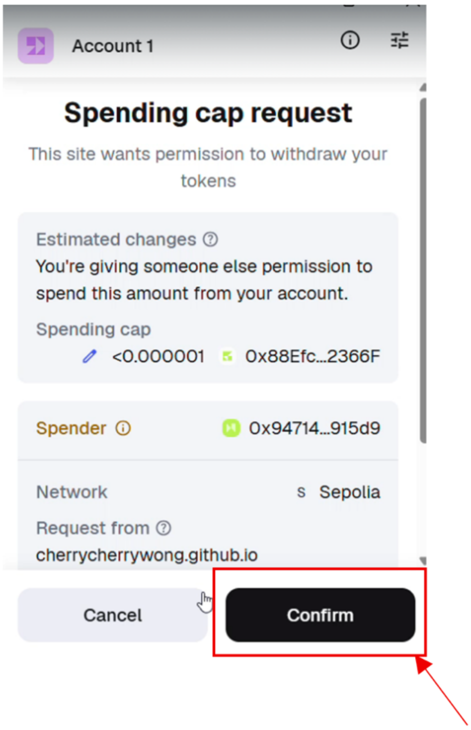

Figure 19 Smart Contract Token Authorization and Spending Cap Request shown in User’s MetaMask Wallet

To maintain transparency during this asynchronous blockchain operation, the Automated Status Synchronization feature (see Figure 20) locks the web interface and provides real-time feedback, notifying the user that the system is "Swapping for ETH..." while the network validates the spending permission.

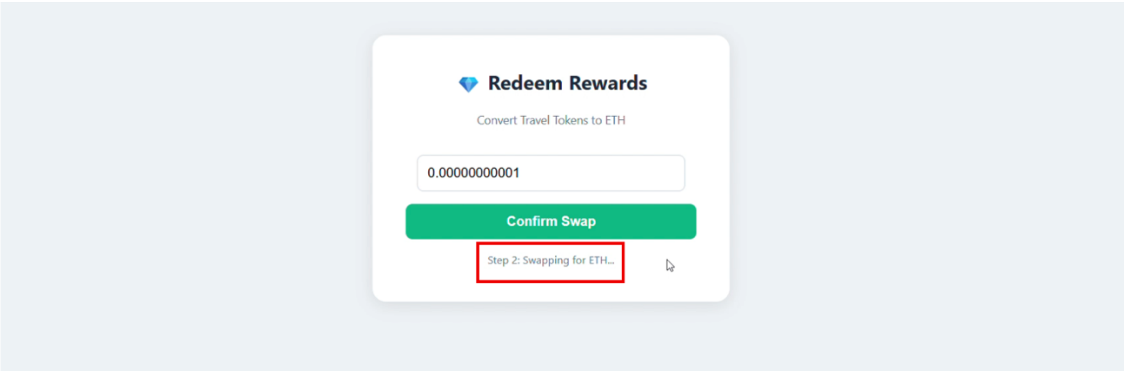

Figure 20 Asynchronous Transaction Status Feedback shown in the DDW after User clicking “Confirm” button in one’s MetaMask Wallet

Following the authorization phase, the system activates the Atomic Swap Confirmation feature. This secondary verification step presents a MetaMask "Transaction request" that outlines the finalized swap parameters, including the Estimated Asset Inflow (with incoming ETH highlighted in green) and the calculated Network Gas Fees.The final stage is the Automated Payout and Settlement feature. Upon selecting "Confirm," the smart contract executes the swap on-chain (see Figure 21).

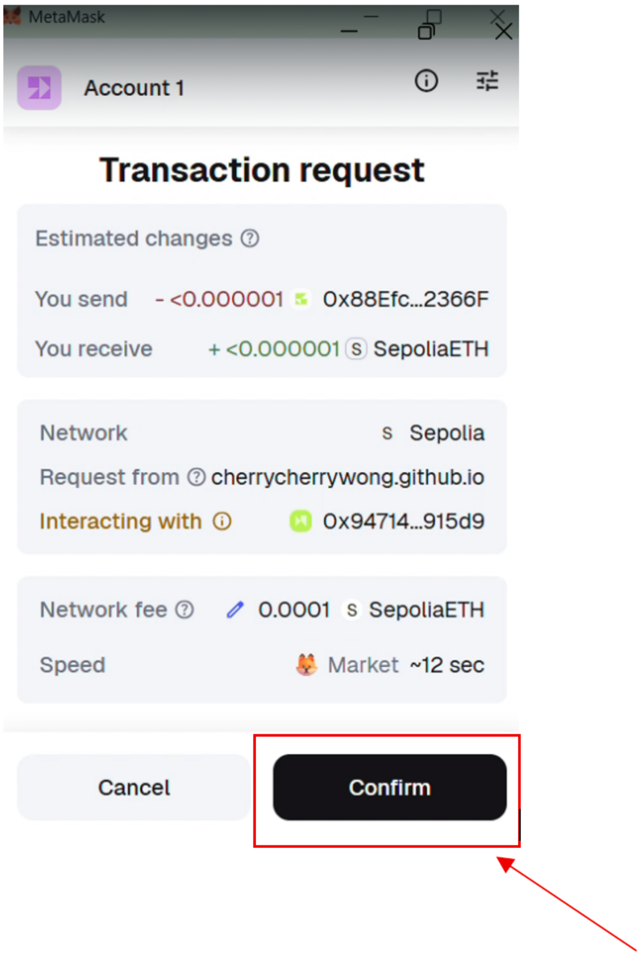

Figure 21 Transaction Request Analytics and Final Swap Confirmation shown in MetaMask Wallet

Immediate confirmation is provided through the On-Screen Success Notification shown in Figure 22, which verifies that the transaction was successfully mined. This completed cycle results in a Real-Time Balance Update, where the contributor's liquid holdings are reflected within their wallet extension as verified ETH.

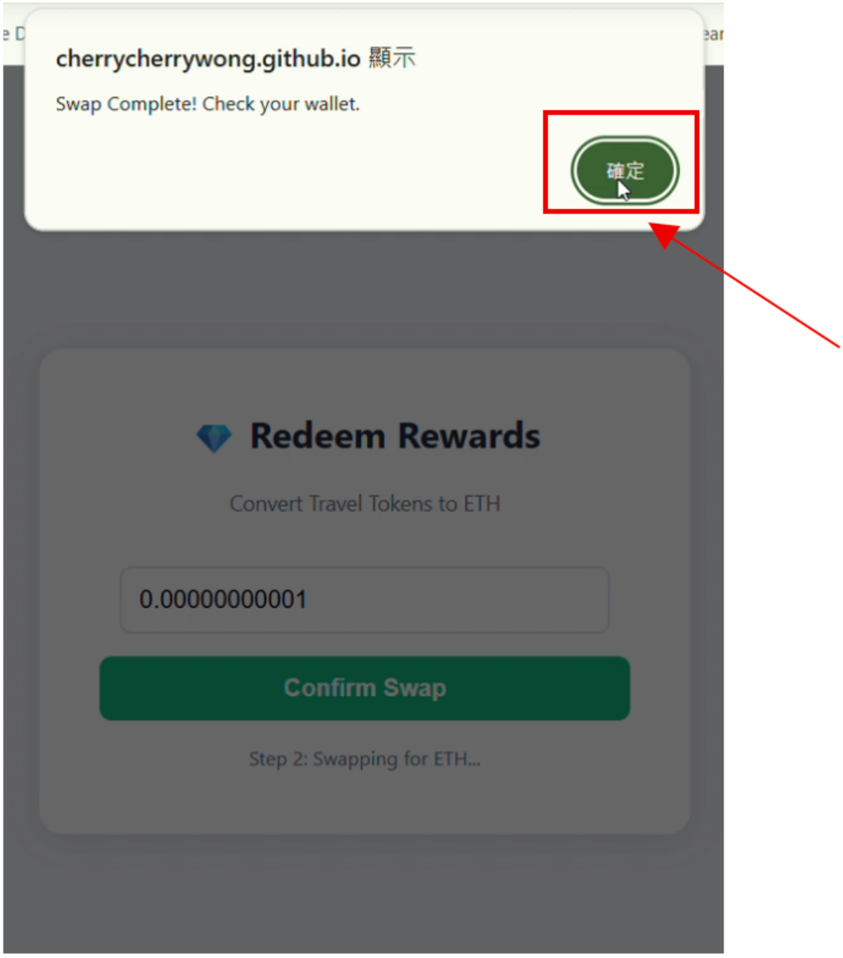

Figure 22 Success Notification upon Successful Redemption of ETH

Connecting to the MetaMask wallet serves to establish a **Secure Financial Bridge** that allows users to redeem platform tokens earned from data contributions back into Ethereum (ETH). This link is essential for the interface to display real-time balances and to authorize the smart contract transactions required for automated reward payouts. Ultimately, this integration ensures that the DDW maintains a trustless, self-sustaining economy where data value is directly and transparently converted into tangible digital assets.

3.4.2 Data Consumers Interface: Marketplace Discovery and Profile Selection

The enterprise entry point into the ecosystem leverages the same secure Unified Login Interface used by data contributors. This ensures a consistent security posture and a frictionless onboarding experience for all network participants, regardless of their role.

Enterprises, as data consumers, begin their interaction at the Social SSO Login screen (see Figure 8) and establish their decentralized identity by performing the MetaMask Cryptographic Handshake (see Figure 10). The transition to the buyer-specific environment occurs when the authenticated user selects the "Data Consumer" profile and triggers the "Browse Marketplace" call-to-action (see Figure 9). This action redirects the user from the general landing hub to the specialized Data Buyer Portal, where they can begin the discovery process (see Figure 23).

Upon entering the portal, consumers are presented with Aggregated Network Statistics that provide a macro-level overview of the available data pool (see the purple box marked in Figure 23).

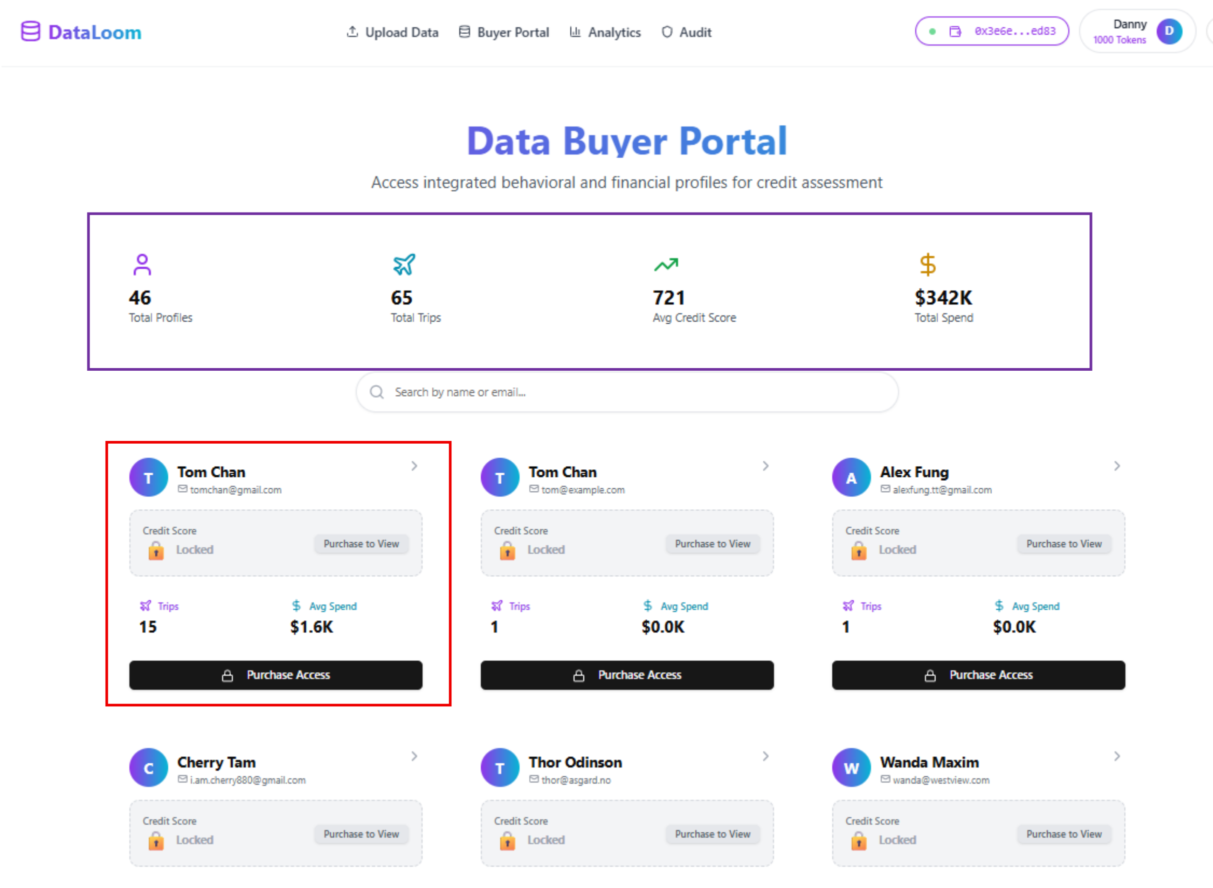

Figure 23 Data Consumer's Portal Interface and Personal Profile Discovery Dashboard

This dashboard serves as a high-level feasibility tool, allowing enterprises to gauge data density through four key metrics:

- **Total Profiles:** The aggregate number of unique personal data profiles available for enterprise access.
- **Total Trips:** The cumulative volume of travel events recorded within the DWW.
- **Avg Credit Score:** A network-wide benchmark indicating the general reliability of the participant pool.
- **Total Spend:** An indicator of the collective commercial potential and economic activity of the available profiles.

To uphold the platform's Privacy-by-Design principles, granular insights, specifically the Alternative Credit Score for case demonstration, remain Locked by default (see the red box marked in Figure 23). This ensures that sensitive behavioral markers are only decrypted and revealed after a verified permissioning transaction.

Next, data consumers can browse and evaluate available data profiles through a grid of anonymous User Cards. This interface allows enterprises to filter and identify relevant behavioural profiles without revealing individual identities. Each User Card represents a distinct personal data profile and displays a set of public metrics. These public metrics displayed on each card allow consumers to evaluate the relevance and quality of a profile prior to purchase.

3.4.3 Interactions between Data Consumers and Data Contributors: Insight Access and Secure Acquisition

The final phase of the marketplace interaction involves a transparent and blockchain-verified protocol that facilitates the exchange of platform capital for granular behavioral insights. This phase ensures that every data acquisition is recorded on a public ledger, providing an immutable audit trail for both the enterprise and the contributor.

Smart contract verification and on-chain transparency underpin the consumer purchase workflow by ensuring that each transaction is executed through a secure and auditable protocol. After identifying a data profile that satisfies their risk assessment criteria, the consumer initiates a purchase request to view the associated data insights (e.g. alternative credit scores).

As illustrated in Figure 24, the interface displays a “Confirm Data Access Transaction” modal, which includes a “View on SepoliaScan” option.

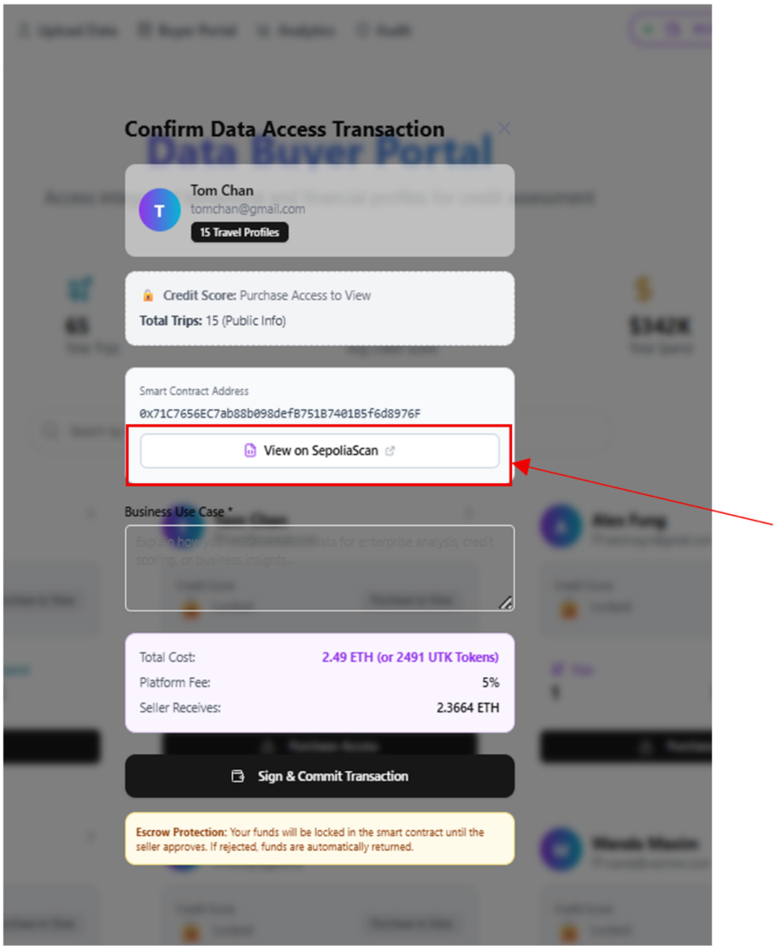

Figure 24 Data Access Confirmation & Smart Contract Gateway shown on the Confirm Data Access Transaction

By clicking the "View on SepoliaScan" button, this allows the consumer to inspect the corresponding smart contract address and the details of the smart contract on the Sepolia Testnet Explorer (see Figure 25). This allows the data consumers to verify the contract source code and confirm the terms of the data exchange are enforced on-chain without reliance on a central intermediary.

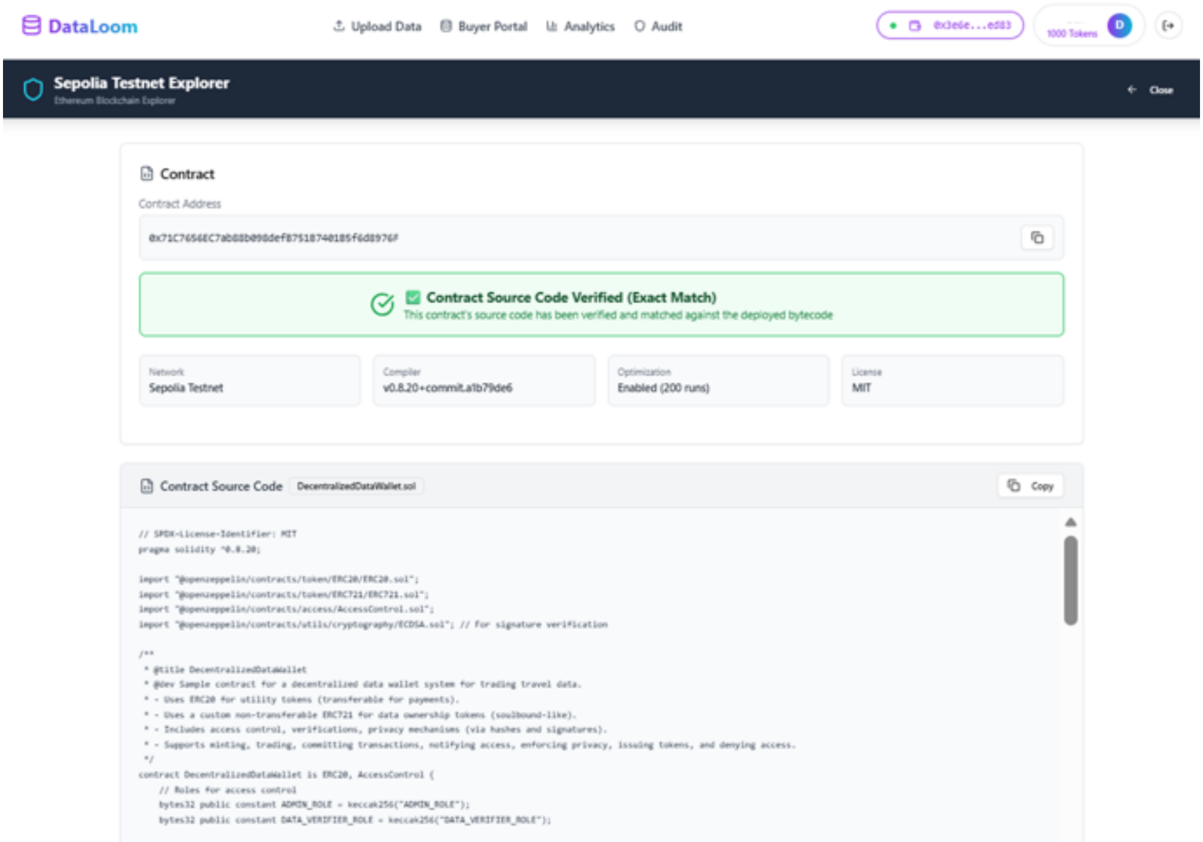

Figure 25 Public Blockchain Audit and Smart Contract Verification Interface shown on Sepholia Testnet Explorer

The Business Use Case Audit and Transaction Cost Review modules are utilized to finalize the purchase while maintaining network-wide accountability. Before committing capital, the consumer must populate the "Business Use Case" field, detailing the professional reason for accessing the profile's private data. This input is recorded on-chain, creating a permanent record of the data's intended use. Simultaneously, the consumer utilizes the Pricing Breakdown feature (see Figure 26) to review the financial parameters, including the Total Cost (e.g., 2.49 ETH), the Platform Fee (5%), and the final net amount the Seller (ie. Data Contributor) Receives.

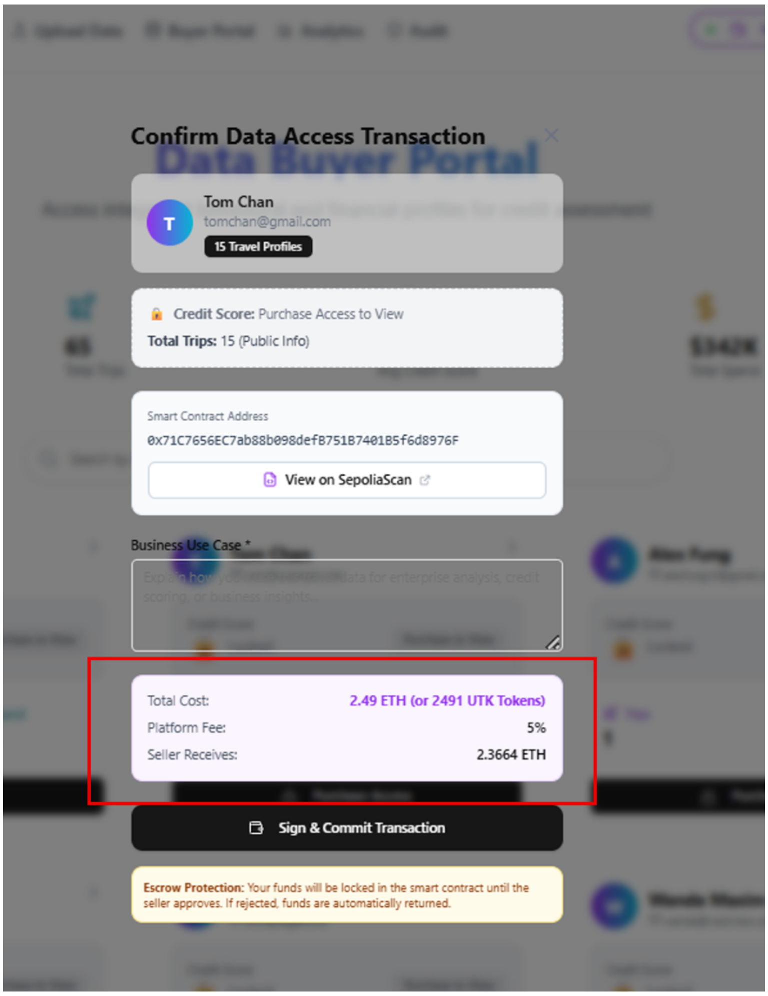

Figure 26 Business Use Case Entry and Transaction Cost Analysis shown on the Confirm Data Access Transaction

Cryptographic Signing and Automated Decryption facilitate the final asset transfer. Triggered by the "Sign & Commit Transaction" action within the buyer's portal (see Figure 27), this feature initiates a MetaMask wallet interaction where the consumer reviews the network gas fees and authorizes the total payment for the profile access.

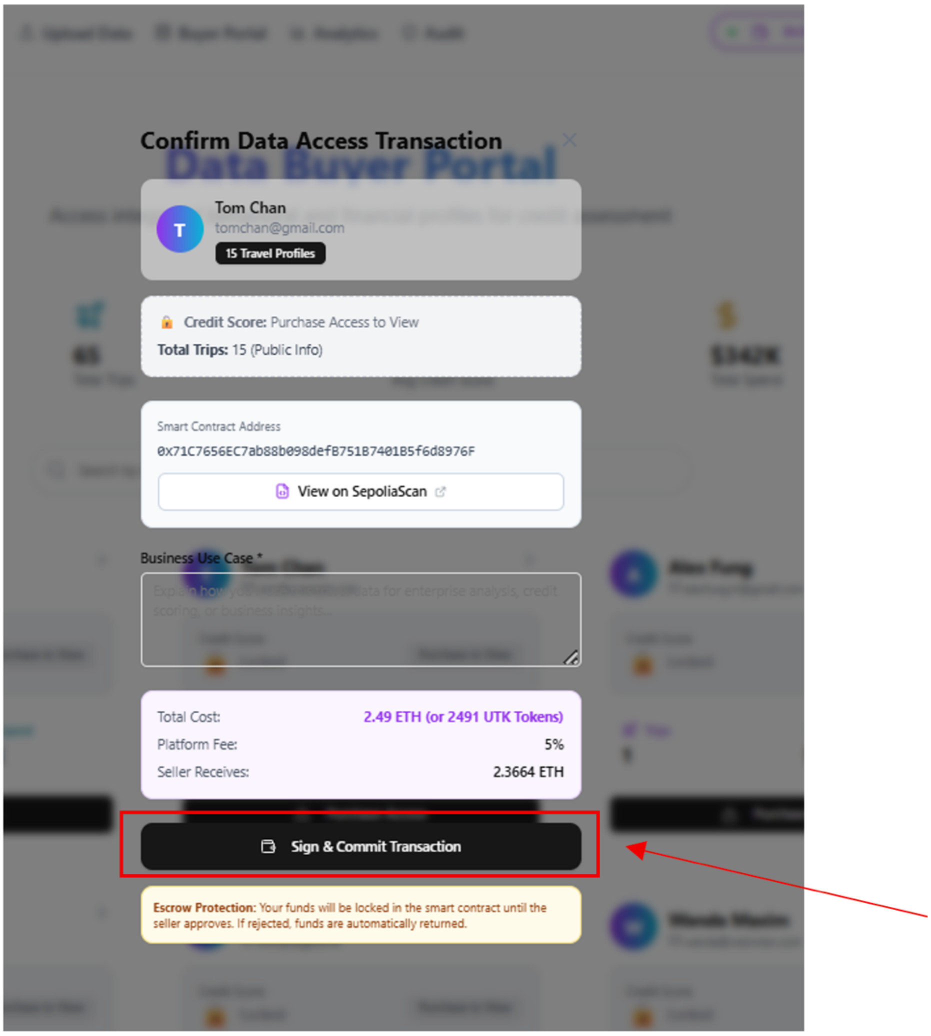

Figure 27 Transaction Execution with Digital Signature and Cryptographic Commitment Action shown on the Confirm Data Access Transaction

To ensure process transparency, the consumer interface provides a Real-Time Transaction Status update, displaying a "Waiting for seller (data contributor's approval)..." notification while the request is broadcast across the network (see Figure 28).

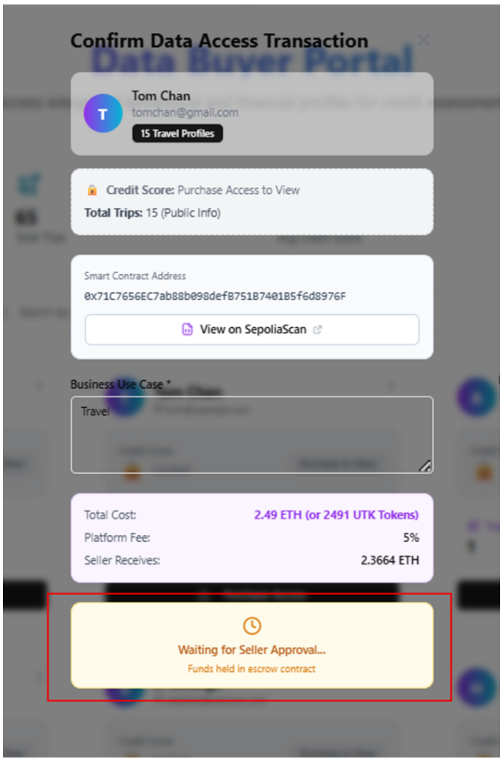

Figure 28 Data Consumer-Side Pending Approval Status Interface shown shown on the Confirm Data Access Transaction

Bidirectional approval and data sovereignty are maintained through a final verification step on the data contributor's interface, ensuring the individual retains ultimate control over their personal assets. As shown in Figure 29, an automated Data Insight Request (New Purchase Request) pop-up appears on the contributor's dashboard, presenting the specific details of the data consumer' request. This allows the data contributor, ie. the individual, to exercise their governance rights by choosing to either Accept or Reject the transaction. Once the contributor grants approval, the smart contract finalizes the atomic swap, triggers the automated decryption protocol, and settles the rewards directly into the contributor's wallet.

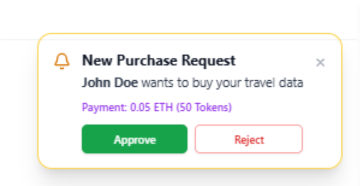

Figure 29 Contributor-Side Data Access Request and Governance Interface showing the Accept and Reject Options

Automated access confirmation and Decryption finalization are triggered the moment the data contributor approves the marketplace request, validating the trustless settlement of the insights-for-payment exchange. As shown in Figure 30, the consumer interface provides immediate visual feedback via a green success checkmark and the notification: "Access Granted! Data unlocked successfully." This feature confirms that the underlying smart contract has verified the transaction and released the cryptographic keys necessary to decrypt the contributor's private behavioral insights.

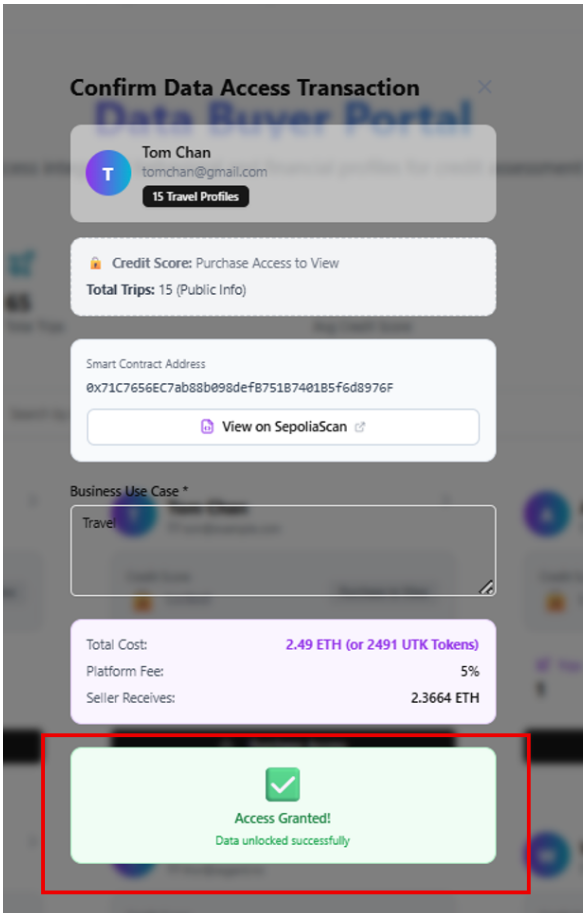

Figure 30 Access Granted and Successful Decryption Notification shown on the Confirm Data Access Transaction

Dynamic Profile State Transition and the Dashboard Refresh Module automatically update the consumer's view to reflect the newly acquired permissions without requiring a manual page reload. Once the data is unlocked, the "Locked" icons on the specific User Card vanish, revealing the contributor's actual Alternative Credit Score. Simultaneously, the call-to-action button undergoes a functional shift, changing from "Purchase Access" to "View Details" (see Figure 31), which serves as the entry point to the granular data environment.

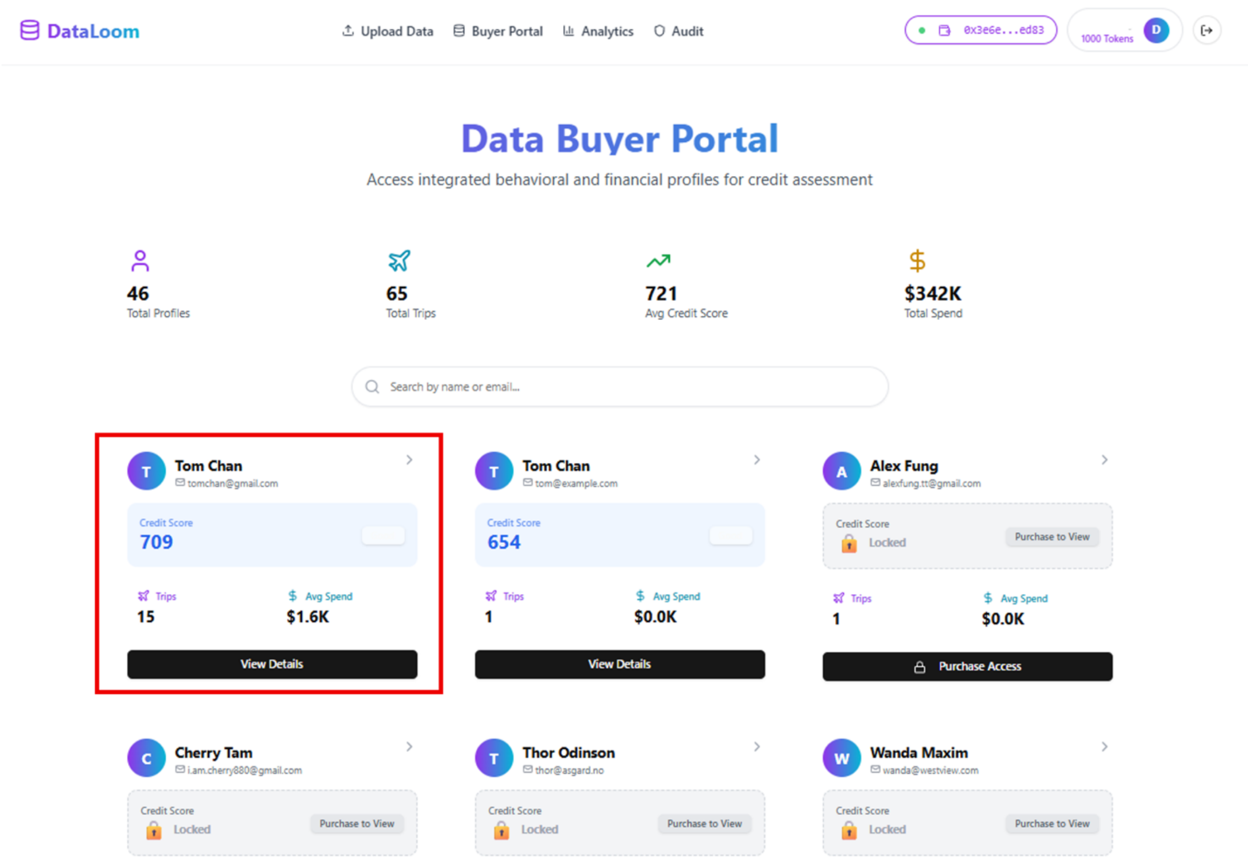

Figure 31 Refreshed User Dashboard with Unlocked Credit Metrics shown on the Data Consumer's Interface

Upon clicking the "View Details" butoon, the Behavioral Data Assessment and Multi-Dimensional Insights Reporting Module presents the final output of the data exchange: a comprehensive "Credit Assessment Report" shown in Figure 32. This feature organizes the contributor's travel-related behavioural metadata into four strategic analytical categories to aid in enterprise decision-making:

1. **Financial Ability & Spending Power:** Aggregates metrics such as average spend per trip and travel class to estimate user liquidity and purchasing power.
2. **Lifestyle Stability Indicators:** Evaluates trip frequency and destination consistency as a proxy for social and professional reliability.
3. **Willingness & Personal Responsibility:** Measures booking lead times and payment timing to assess organizational habits and financial discipline.
4. **Recent Trip History:** Provides a transparent, scrollable audit trail of specific past travel events that support the aggregate scores, ensuring data provenance.

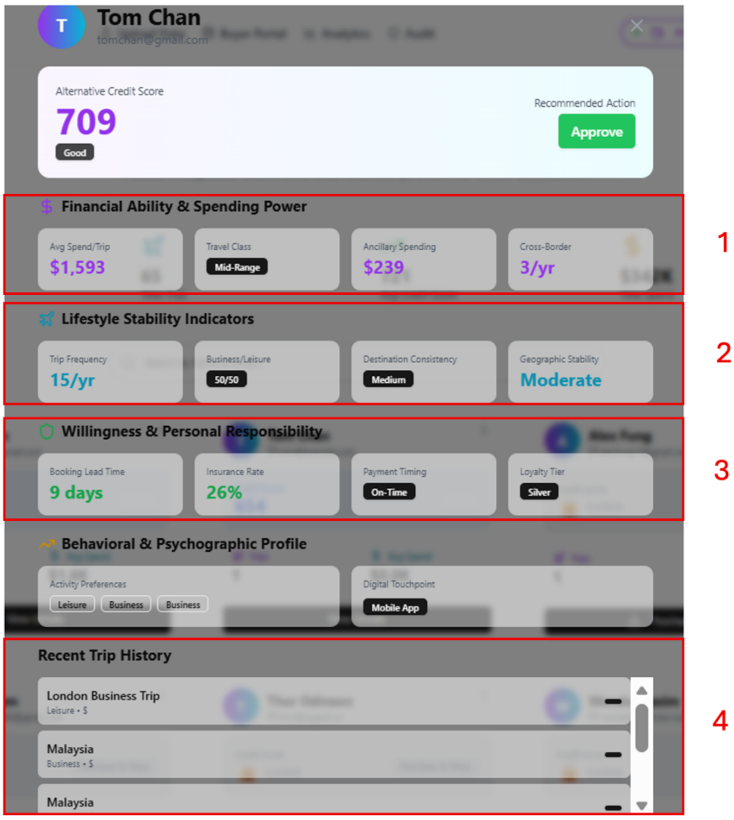

Figure 32 Comprehensive Behavioral Insights and Credit Assessment Report of the Data Consumer – "Tom Chan"

3.5 Backend Implementation of ML Algorithm Logic for DPI Generation

Section 3.5 describes the backend implementation of the DPI Generation Engine, which transforms verified behavioural data into actionable insights. Section 3.5.1 explains the selection of the Random Forest model over alternative algorithms, emphasizing predictive accuracy, interpretability, and computational efficiency. Section 3.5.2 details the feature engineering pipeline, including the extraction, transformation, and preprocessing of raw behavioral signals into structured input features for the ML engine. Finally, Section 3.5.3 covers insight generation and alternative credit score synthesis, describing how the model produces DPI and mathematically derives alternative credit scores while preserving contributor privacy and compatibility with the on-chain architecture.

3.5.1 Algorithmic Selection and Comparative Model Evaluation

The DPI Generation Engine utilizes a sophisticated machine learning framework to transform heterogeneous behavioral data, retrieved directly from user-uploaded Data Profiles, into verifiable risk metrics and actionable insights. As established in the interim report, this engine is specifically engineered to process travel-related behavioral metadata to generate the Alternative Credit Score alongside supplementary qualitative insights, effectively converting raw contributor data into high-value financial intelligence.

The **Random Forest Classifier** was selected as the optimal ensemble learner following a rigorous evaluation of multiple architectural alternatives, specifically to address the non-linear and stochastic nature of contributors' travel-related behavioral data while ensuring the engine remained both accurate and economically viable for a decentralized ecosystem.

First, **linear and logistic regression models** were excluded because they rely on first-order linear dependencies and cannot capture the complex, multivariate interactions present in discrete behavioral signals, such as the non-linear relationship between spending volatility and profile metadata.

Second, **single decision trees** were rejected due to their high variance and tendency to **overfit**, resulting in poor generalization on out-of-sample data which would compromise the integrity of the credit score.

Third, **K-Nearest Neighbors (KNN)** was disqualified primarily because of high computational latency during real-time API polling and its sensitivity to the “curse of dimensionality” when processing high-feature vectors.

Fourth, **neural networks**, despite their strong predictive capabilities, were dismissed owing to their “black box” nature and extreme computational inefficiency. Moreover, given the platform’s roadmap toward potential on-chain execution, the millions of matrix multiplications required for neural network inference would incur prohibitive gas costs and exceed block gas limits.

In contrast, the **Random Forest model** provides a more scalable and interpretable approach, offering a feasible path for decentralized verification while maintaining the high predictive performance required for financial decisioning.

3.5.2 Backend Pipeline ML Logic and Alternative Credit Score Synthesis

The backend operationalizes the "Data-to-Asset" pipeline through a structured sequence of model training and real-time feature engineering. The predictive engine was developed using a supervised learning pipeline trained on 43 synthetic data records designed to simulate real-world travel and expenditure patterns. To ensure the model is robust against outliers and noise, we utilized an ensemble of 100 decision trees, where the final prediction is determined through a majority voting mechanism. This approach, known as Bootstrap Aggregating (Bagging), is a standard industry practice for credit risk modeling as it reduces variance and improves model stability compared to single-tree architectures [21].

The system extracts **three primary latent features** to serve as proxies for creditworthiness, following the "Ability, Stability, and Responsibility" framework common in alternative lending [22]:

1. **Trip Frequency:** Representing temporal stability and lifestyle consistency.
2. **Mean Expenditure per Trip:** Approximating disposable income and repayment capacity.
3. **Profile Completeness Index:** Quantifying user reliability and transparency in data disclosure.

During the Real-Time Inference stage, the system produces a predicted probability (x) between 0 and 1, representing the statistical likelihood of a user being a "Good Payer." To make this actionable for financial institutions, a Linear Normalization technique is applied to map this probability to a standard industry range established by the FICO Score (introduced in 1989) and VantageScore 3.0 (2013) [23].

The transformation is governed by the following linear equation:

$$y = 300 + 550x$$

Where:

- y: The final **Alternative Credit Score**.
- x: The **Predicted Probability** generated by the Random Forest model ($0 \leq x \leq 1$).
- 300: The **Intercept**, representing the minimum possible industry score.
- 550: The **Slope**, representing the total point spread (scaling the $0 - 1$ range to a 550-point spread).

3.5.3 Risk Categorization and Leveling

To ensure interoperability with enterprise underwriting systems, the model applies a deterministic risk stratification function that converts the final alternative credit score into predefined risk tiers (see Table 5). Each tier corresponds to explicit decision thresholds used in automated credit evaluation pipelines, enabling lenders to directly operationalize the score within existing approval, pricing, and risk-control frameworks [44].

Risk Tier	Score Range	Behavioral and Financial Implication
Excellent	≥ 800	Exceptional financial capacity and superior behavioral consistency
Very Good	$740 - 799$	High reliability with minimal observed risk markers
Good	$670 - 739$	Moderate reliability and stable, consistent behavioral patterns
Fair	$580 - 669$	Average reliability; may exhibit minor inconsistencies in data or spending
Poor	≤ 580	High risk; signifies significant volatility or low profile transparency

Table 5 Alternative Credit Risk Stratification Tiers

Note: This table provides a standardized mapping between the numerical alternative credit score and corresponding risk categories, detailing the behavioral and financial implications of each tier.

By combining the predictive depth of the 100-tree Random Forest model with this deterministic Risk Leveling framework, the DDW backend delivers a transparent, multi-dimensional assessment of risk that is both mathematically sound and ready for enterprise-level deployment. This ensures that the generated DPI are not just raw numbers, but actionable intelligence for real-world financial decisioning.

3.6 API Operational Architecture and Interface

To facilitate seamless integration with enterprise systems, the platform features a custom-developed API Gateway that serves as the central bridge between the DDW and external financial evaluators.

As shown in the Alternative Credit Scoring API Initiation Interface (see Figure 33), the system provides a streamlined portal through which enterprise consumers can initiate inquiries using a contributor's unique identifier. This action triggers a real-time, asynchronous API call that coordinates the entire data retrieval and machine learning pipeline, including the aggregation of behavioral metadata, execution of Random Forest inference, and normalization of the resulting score, all within a single low-latency transaction. By consolidating these complex backend processes into a unified gateway, the platform delivers a "plug-and-play" solution that enables lenders to access verifiable behavioral insights and standardized alternative credit scores without managing the underlying blockchain or ML infrastructure.

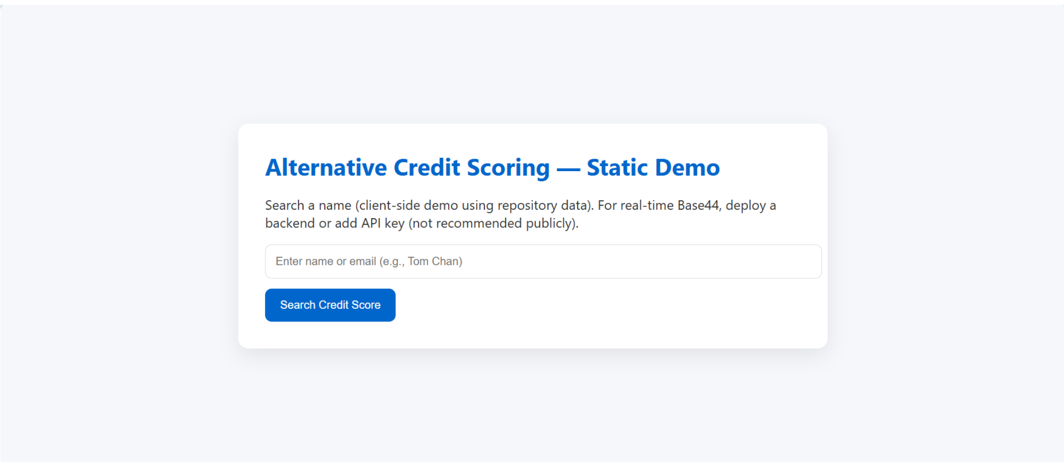

Figure 33 Alternative Credit Scoring API Initiation Interface

3.7 Limitations & Mitigations

A key limitation of the initial DDW model is its reliance on user-uploaded data, which introduces the risk that contributors could falsify or alter travel records to influence their alternative credit outcomes. A growing trend in the industry addresses this challenge by shifting toward a source-to-chain model, where credentials, such as purchase history, or financial transaction data, academic records, utility bills, are issued directly onto the blockchain by the trusted source institution rather than by the individual contributor. When data is anchored on-chain by a verified issuer, it becomes inherently authenticated: the issuer's cryptographic signature confirms the origin, and the immutability of the blockchain ensures the data has not been modified since issuance [25].

This approach is already implemented in the academic sector, with institutions such as MIT and the University of Nicosia issuing blockchain-based digital diplomas that cannot be falsified by recipients [26]. Similarly, in the high-value asset sector, companies like Everledger utilize blockchain to record the provenance and ownership of luxury goods—such as diamonds and fine wine, directly at the point of origin or sale [27]. By adopting a similar methodology, the DDW platform can transition from manual verification processes to a trustless, automated environment. In this envisioned state, behavioral data would be verified by design upon entry to the blockchain, ensuring that the machine learning engine operates exclusively on legitimate, tamper-proof information from the outset [28].

3.8 Current Status & Future Work

The DDW prototype has reached an early stage where results are acceptable and align with the project schedule, meeting initial expectations. Ongoing work focuses on continuous refinement of the prototype, particularly the insight generation engine, and preparing for subsequent phases, including alpha and beta testing and interviews with industry practitioners, planned for the next semester.

Looking ahead, the primary objective is to explore on-chain machine learning, executing algorithms directly within smart contracts to enhance decentralization. This may involve simplifying existing models, such as the Random Forest used for alternative credit scoring, to ensure feasibility within the EVM's computational limits. In parallel, the platform aims to develop one or two additional lightweight use cases, such as health and wellness analytics, where simple ML algorithms can be executed on-chain to demonstrate the DDW’s versatility.

Furthermore, in the upcoming semester, we will attempt to adopt real-world data during the testing phases to gradually replace synthetic data within the data upload module. The system will undergo continued modification to accommodate a broader range of data domains and modalities, ensuring more robust feature engineering for ML training and mitigating the risks of data mismatch between simulated and live environments.

As an alternative approach, if full on-chain ML proves computationally prohibitive, efforts will focus on advancing off-chain ML workflows. This includes refining feature selection, integrating open-source real-world datasets for improved feature engineering, and fine-tuning the model to increase the predictive accuracy of the alternative credit scores. Additionally, continuous improvements to the user interfaces will be undertaken to ensure a seamless and intuitive experience for both data contributors and enterprise consumers, preparing the platform for broader testing and adoption.

4. Conclusion

The DDW project was initiated to address a structural imbalance in the digital economy, namely the erosion of personal data sovereignty and the absence of transparent mechanisms through which individuals can derive value from their own data. The primary objective of the project was to investigate whether a blockchain-integrated architecture could securely transform raw behavioral metadata into verifiable and monetizable digital assets. To achieve this objective, a four-layer system was developed, spanning a React-based frontend, a backend machine learning pipeline built around a Random Forest ensemble, and smart contract deployment on the Sepolia testnet, collectively demonstrating a complete and privacy-preserving “data-to-asset” lifecycle. The project’s current progress validates the technical feasibility of the DDW framework through the successful implementation of dual-facing frontend interfaces designed for both data contributors and data consumers. On the backend, the ML generation engine utilizes a Random Forest algorithm ensemble to synthesize heterogeneous behavioral signals into a standardized Alternative Credit Score, while a custom API gateway ensures that these decentralized insights are delivered to enterprise users in an actionable and near real-time format. Collectively, these outcomes contribute a functional blueprint for the emerging tokenized data economy, demonstrating that blockchain technology serves as more than a currency ledger by functioning as a trust infrastructure for data provenance, consent enforcement, and value exchange. By bridging decentralized user profiles with enterprise decision-making workflows, the study provides key insights into how individuals can reclaim the economic value of their digital footprints without compromising privacy.

Despite these milestones, the prototype currently exhibits a data provenance gap, as reliance on manual data uploads remains susceptible to fraudulent manipulation. Furthermore, due to computational constraints within the Ethereum Virtual Machine, machine learning inference is presently executed off-chain. An additional limitation arises from the use of synthetic travel-related behavioral data to train the machine learning model for alternative credit score generation, as such data lacks the complexity and stochastic volatility inherent in real-world global travel behaviors. This constraint may limit the granularity and predictive accuracy of the generated insights, as model performance remains bounded by simulated parameters rather than the nuanced and unpredictable dynamics observed in live financial ecosystems.

Future research should therefore prioritize a transition toward a “source-to-chain” model, in which trusted institutions anchor data directly to the blockchain to enable automatic verification. In parallel, the integration of zero-knowledge proofs (ZKPs) should be explored to support privacy-preserving on-chain machine learning inference without exposing underlying feature vectors. These enhancements are critical for evolving the DDW from a functional prototype into a mature, scalable, and privacy-first platform suitable for real-world deployment.

5. Project Schedule

Time Period	Tasks	Completion
September	• **Meet with supervisor**	**Done**

	• ***Phase 1 Deliverable***
	    o ***Detailed project plan***
	    o ***Project website***
		Done

	• Do literature reviews on tokenization and tokenomics in P2P data ecosystems, blockchain protocols and architecture, privacy-preserving techniques, and alternative credit scoring models • Identify suitable behavioural data-specific domain, modalities, and their derived insights	**Done**

October	• **Case studies review:**
	    o Decentralised P2P data marketplace
	    o Self-sovereign identity specifications
	    o Decentralised AI machine training
	• **Refine review questions based on MLR findings**
	• **Develop and draft technical protocol**
	    o **Define data types, token standards, access control logic and layered architecture**
		Done

November	• **PoC Development (Phase 1): frontend interface**	**Done**

December	• **PoC Development (Phase 2): system architecture**
	    o Backend logic
	    o Blockchain integration
	    o Smart contract development
		Done

January	• **PoC Development (Phase 3): integration** o Connect backend with frontend • **Prepare for first presentation**	**Done**
	• ***Phase 2 Deliverable*** o ***Interim Report***	Done

February	• **PoC Development (Phase 4): On-chain ML**
	    o Connect backend with frontend
	• **Internally Technical & Operational Evaluation**
	• **Conduct Beta testing**
	• **Collect Beta testing Feedbacks**
	• **PoC Development (Phase 4): refinement after beta testing**
	• Evaluation on practicality & limitations (Phase 1):
	    o Analyze quantitate metrics
	    o Analyze beta testing feedbacks
		In Progress

March- April	• Evaluation on practicality & limitations (Phase 2):
	    o Get insights industry practitioners
	• **Write final report**
	    o **Prepare for final presentation**
	    o **Finalise project website with PoC demo**	

April	• ***Phase 3 Deliverable*** o ***Final report***	

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